

Money, Credit, and Banking in a Heterogeneous-Agent New-Monetarist Model*

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Abstract

We revisit the liquidity-transformation mechanism of competitive banking in Berentsen, Camera and Waller (2007), a representative-agent economy, and study how its implications change in an empirically relevant heterogeneous-agent environment. Unlike the representative-agent model, our model endogenously generates a positive mass of hand-to-mouth households with zero liquid assets, leading to a right-skewed distribution of liquidity. In this setting, bank credit substitutes for cash self-insurance and reduces equilibrium money demand despite the presence of deposit returns. Quantitatively, banking is still essential, but the source of these gains differs from Berentsen et al. (2007) and varies with inflation: welfare improvements are relatively more deposit-driven at low inflation, while they become increasingly borrower-driven at higher inflation.

JEL codes: E31; E42; E51; E52.

Keywords: Money, Credit, Banking, Heterogeneity, Welfare, Inflation

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1 Introduction

This paper revisits the core mechanism of [Berentsen et al. \(2007\)](#) by re-examining the liquidity-transformation role of competitive banking and its welfare implications. Our objective is to understand how banking affects the distribution and use of liquid assets when agents face heterogeneous liquidity needs and trading frictions. In doing so, we study how competitive banking shapes consumption behavior, the emergence of hand-to-mouth households, and the welfare consequences of inflation.

To address these questions, we embed the competitive banking structure of [Berentsen et al. \(2007\)](#) into a heterogeneous-agent New Monetarist framework ‘à la [Rocheteau, Weill and Wong \(2021\)](#). [Berentsen et al. \(2007\)](#) (hereafter, BCW) provide a tractable representative-agent model in which banks offer credit and deposit contracts but distributional forces are absent. [Rocheteau et al. \(2021\)](#) (hereafter, RWW), in contrast, generate a non-degenerate money distribution through inelastic labor supply and idiosyncratic trading shocks, thereby producing heterogeneity in money holdings and consumption. By integrating BCW’s competitive banking into the heterogeneous monetary environment of RWW, we develop a model that captures both the liquidity-transformation function of banks and the distributional margins that arise from heterogeneous money balances. This synthesis enables us to analyze how banking reallocates liquidity across agents and how these reallocations shape consumption, money demand, and welfare.

Our contribution is to examine competitive banking in a heterogeneous-agent monetary economy, rather than in a representative-agent setting as in BCW. This approach reveals distributional and welfare implications that are invisible in the original BCW framework. Specifically, we show that banking’s liquidity transformation reshapes the distribution of liquid-asset holdings and drives the endogenous emergence of hand-to-mouth households who optimally choose to hold zero money balances. Moreover, the welfare effects of inflation differ sharply from those in BCW: borrowers and depositors face systematically different inflation burdens, producing an inflation-dependent pattern of welfare incidence across household types. These insights provide a new perspective on the role of competitive banking in monetary economies with heterogeneity.

Our analysis delivers two central theoretical implications that sharply distinguish the heterogeneous-agent economy from the representative-agent benchmark of BCW. First, once liquidity heterogeneity and endogenous hand-to-mouth behavior are introduced, competitive banking reduces aggregate money demand in equilibrium. While deposit services raise the return to holding liquidity, credit access relaxes the need for precautionary cash self-insurance at the lower tail of the distribution and generates a positive mass of households that optimally choose zero real balances. The resulting leftward shift in the stationary distribution compresses average money holdings and shrinks the inflation-tax base. Second, the welfare gains from banking become inflation-dependent in composition. At low inflation, gains are relatively more deposit-driven, reflecting improved intertemporal insurance in non-trade states. As inflation rises, however, the dominant welfare margin shifts toward borrowers: credit increasingly substitutes for costly cash self-insurance, and the distributional headwind associated with liquidity reweighting weakens. In contrast to the representative-agent logic, where deposit services summarize the welfare role of banking, our framework implies that the aggregate relevance of banking is ultimately borrower-driven at sufficiently high inflation.

Money, Credit, and Banking in the Heterogeneous Economy We generate heterogeneity in money holdings by assuming fixed labor supply and fixed labor income, so households cannot fully undo idiosyncratic liquidity shocks through labor adjustment and instead accumulate different real balance positions over time. The environment follows the Lagos–Wright framework with decentralized trade and centralized settlement: in the decentralized market (DM), informational frictions make liquid payments essential, while in the centralized market (CM) households work, settle positions, and choose how much liquidity to carry forward. Households face idiosyncratic opportunities to consume in the DM, which create fluctuating liquidity needs across households and periods. Unlike the baseline Lagos–Wright setting—where quasi-linearity and an elastic labor margin eliminate wealth effects and collapse the cross-sectional distribution—fixed labor income reintroduces wealth effects: the marginal value of an additional unit of liquidity depends on the household’s balance-sheet position. Consequently, households’ optimal saving and spending decisions differ across liquidity states, and the stationary distribution of real balances becomes non-degenerate, as in heterogeneous-agent extensions such as RWW.

Building on this heterogeneous liquidity environment, we incorporate a perfectly competitive banking sector as in BCW. After households learn whether they have a consumption opportunity, they can borrow from banks or deposit their balances: households with an opportunity borrow to finance purchases when their liquid wealth is insufficient, whereas households without an opportunity deposit their balances and earn interest before choosing consumption and next-period liquidity. Perfect competition implies zero profits and an equalization of borrowing and deposit rates, so banking reallocates liquidity across households and across states—providing credit when liquidity is scarce and a return when liquidity is idle.

Liquidity intermediation through banking, together with concave utility, delivers an empirically relevant liquid-asset distribution that is sharply right-skewed and features a positive hand-to-mouth mass. With concave utility, liquidity-poor households place a high marginal value on current consumption, making it costly to carry liquidity into the next period. Credit access prevents the marginal value of liquidity from exploding at zero by allowing households with no cash to finance purchases through borrowing. As a result, it is optimal for a positive measure of households to choose zero next-period balances, while liquidity-rich households continue to hold substantial balances to benefit from deposit returns, generating a thin but persistent upper tail. Compared with RWW, where linear utility is often assumed for tractability, strict concavity strengthens balance run-down among liquidity-poor households and helps produce an endogenous mass at zero alongside a right tail, consistent with evidence from the U.S. Survey of Consumer Finances.

Unlike the representative-agent logic in BCW, competitive banking reduces equilibrium money demand in our heterogeneous-agent economy. In the representative-agent benchmark, deposit returns primarily strengthen incentives to carry liquidity across periods, pushing money demand up. In our model, by contrast, credit substitutes for cash self-insurance at the lower tail and creates a sizable mass of households holding zero liquid balances, shifting the cross-sectional distribution left and compressing average real balances despite the presence of deposit interest. Moreover, because lump-sum transfers are proportional to the inflation-tax base, lower aggregate real balances mechanically shrink the equilibrium transfer, adding a general-equilibrium feedback that matters for consumption and for the distribution of inflation burdens across borrowers and depositors.

Insurance and Distributional Effect These money-demand and distributional implications translate into welfare effects that differ sharply from the representative-agent logic in BCW. In our framework, banking affects welfare through two distinct channels: an *insurance effect* and a *distributional effect*, and the balance between them varies systematically with monetary policy (inflation).

The *insurance effect* captures how banking changes allocations conditional on a household's initial liquidity position. Credit allows liquidity-poor households to finance purchases when liquidity is scarce, while deposit services provide a return on idle balances and thus alter saving incentives. Holding fixed the cross-sectional distribution of liquidity, this channel measures the direct, state-contingent improvement (or deterioration) in within-household resource allocation induced by access to borrowing and deposits.

The *distributional effect* captures how banking reshapes the stationary cross-sectional distribution of liquidity and thereby reweights liquidity states in aggregate welfare. By relaxing the need for cash self-insurance, credit can shift households toward lower liquidity positions and generate an endogenous mass at (or near) zero balances. Even if banking improves allocations conditional on liquidity states for some households, the equilibrium reweighting toward low-liquidity states can reduce average welfare, as it places greater weight on states associated with low consumption levels and high marginal utility.

Quantitatively, under a U.S. calibration, banking remains essential: the aggregate welfare gain from banking is positive, and it increases with inflation. The welfare gain is positive because the insurance effect dominates the distributional effect in the aggregate, which implies that the net effect on welfare remains positive even when banking shifts the liquidity distribution to the left. The welfare gain from banking increases with inflation because the negative effect of distributional changes becomes less pronounced at higher inflation. As the self-insurance households without banking reduce liquidity holdings to protect themselves from higher inflation, the distributional effect of banking decreases as inflation rises.

However, the composition of welfare gains changes with inflation in our heterogeneous-agent economy: at low inflation the welfare gain is primarily depositor-driven, whereas at higher inflation it becomes increasingly borrower-driven.

At sufficiently low inflation, welfare gains from banking are mainly driven by depositors rather than borrowers. Although the borrower-side insurance gain from credit is positive, banking simultaneously shifts the stationary liquidity distribution toward lower balances and shrinks the equilibrium transfer through a smaller inflation-tax base. At low inflation, these adverse general-equilibrium forces are large enough to offset the direct insurance benefit of credit for liquidity-poor households in welfare. Deposit services, by contrast, raise the return on idle balances and improve insurance in states in which households do not spend, generating positive depositor-side gains.

On the other hand, as inflation rises, the source of welfare gains shifts from deposits toward credit. Higher inflation increases the opportunity cost of carrying liquidity, making credit a more valuable substitute for cash self-insurance and strengthening the borrower-side insurance benefit. In addition, the self-insurance benchmark also economizes on liquidity at higher inflation, which narrows the gap in stationary distributions and weakens the distributional effect. Depositor-side gains diminish because higher deposit returns are increasingly absorbed by saving choices rather

than by higher contemporaneous consumption, and also because the contraction in real balances and transfers further compresses resources for high-balance households. The net effect is that welfare gains become increasingly borrower-driven at higher inflation.

In sum, while representative-agent analyses like BCW emphasize deposit services as the source of welfare gains from competitive banking, our heterogeneous-agent economy implies that the key margin depends on inflation: deposit-related gains are more relevant at low inflation, whereas credit-related gains dominate at higher inflation. This inflation-dependent composition reflects the interaction of state-contingent insurance effect with endogenous distributional effect, a channel that is absent in representative-agent environments.

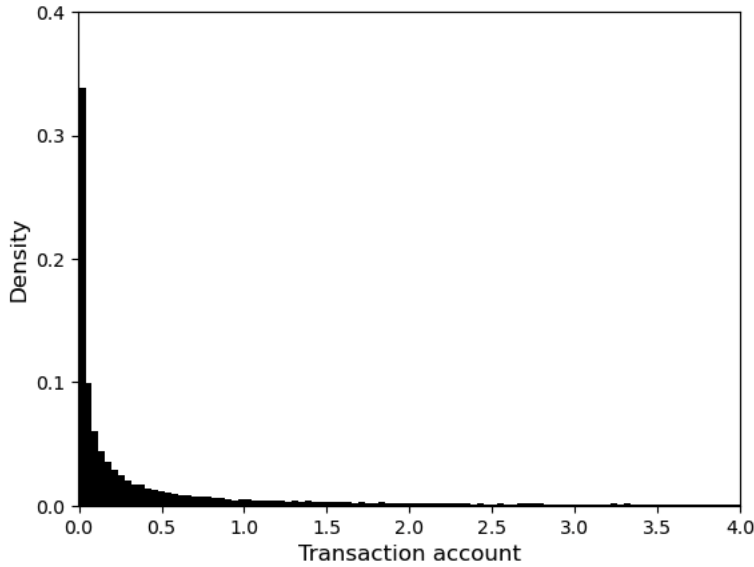
1.1 Motivating Empirical Evidence and Literature Review

Motivating Empirical Evidence A key empirical feature of household liquidity in the United States is that it is both highly unequal and concentrated near zero. Using microdata from the Survey of Consumer Finances (SCF), 1989–2019, we document two robust stylized facts about liquid wealth. First, a non-trivial fraction of households holds very little liquid wealth in transaction accounts, consistent with the presence of an economically meaningful hand-to-mouth (HtM) margin. Second, the distribution of liquid balances is strongly right-skewed: most households hold modest balances, while a small upper tail holds disproportionately large amounts. These facts imply that the marginal value of liquidity is sharply heterogeneous across households and that aggregate money demand cannot be understood through representative-agent logic alone.

To measure liquid wealth, we focus on balances in transaction accounts (e.g., checking and other accounts used for payments), which are directly available for immediate spending and thus provide a natural proxy for transactional liquidity. Figure 1 plots the cross-sectional distribution of transaction-account balances in the SCF and illustrates the two patterns clearly: substantial mass near zero and a long right tail. Importantly, these distributional features are not merely descriptive. They discipline which mechanisms are quantitatively plausible in monetary models. In an economy where many households operate close to a liquidity constraint, policies or institutions that affect the return on liquid assets, access to credit, or the equilibrium inflation-tax base can have strongly state-dependent and distribution-dependent welfare consequences.

These facts motivate our modeling strategy. We study banking in a setting where the distribution of liquid wealth (money) is an endogenous state variable, and where an HtM mass and a right-skewed money distribution arise in equilibrium. This is crucial for revisiting the classic “why banking?” question. In the representative-agent banking benchmark of [Berentsen et al. \(2007\)](#), heterogeneity in liquidity positions is absent by construction, so the model cannot speak directly to (i) who is liquidity constrained, (ii) how banking reshapes the cross-sectional incidence of the inflation tax and transfer rebate, or (iii) how credit access interacts with an uneven liquidity distribution. In contrast, once the empirically salient heterogeneity is built in, banking can remain essential for welfare, but the source and composition of welfare gains need not mirror the representative-agent mechanism: deposit returns, borrowing opportunities, and distributional reweighting can interact in ways that are invisible in a degenerate-liquidity economy.

Figure 1: Distribution of money holding in the SCF



Note: This figure displays the distribution of transaction account balances across U.S. households, based on data from the Survey of Consumer Finances (SCF) spanning from 1989 to 2019. The value of each transaction account has been normalized by dividing it by the average value, and the data for the top 5% segment has been truncated.

Literature review We study a classic question in monetary theory: when, and why, is banking essential for welfare? A leading benchmark is [Berentsen et al. \(2007\)](#), which provides a tractable representative-agent framework in which competitive banks are welfare-improving primarily because deposit services raise the return to holding liquidity and thereby strengthen intertemporal insurance; in that setting, the welfare gain from banking is largely deposit-driven and increases with inflation over a relevant range. Our analysis revisits this benchmark in an empirically relevant heterogeneous-agent environment with a non-degenerate liquidity distribution and an endogenous hand-to-mouth margin. In such an environment, banking remains essential, but the source of its welfare relevance can differ: credit access and distributional reweighting interact with deposit returns, so the composition of welfare gains need not mirror the representative-agent mechanism emphasized in BCW.

Recently, [Kam, Lee, Lee and Ng \(2025\)](#) study competitive banking in an economy that is ex ante symmetric but ex post heterogeneous, where a non-degenerate distribution of posted prices sorts buyers into money users, credit buyers, or depositors. Their quantitative results emphasize a trade-off between liquidity-insurance benefits and an adverse price-dispersion/market-power channel, and show that the aggregate welfare gain from banking is driven mainly by credit buyers. Our environment differs in two key respects. First, we generate ex ante heterogeneity through fixed labor income and idiosyncratic spending opportunities, yielding a non-degenerate stationary distribution of liquid wealth rather than heterogeneity driven primarily by realized prices. Second, banking affects equilibrium money demand and the inflation-tax base via endogenous liquidity accumulation and distributional reweighting, so the welfare effects of inflation reflect the interaction between an insurance channel and a distributional (reweighting) channel, without relying on

endogenous price dispersion.

Our paper relates to a large literature in heterogeneous monetary economics that studies how trading frictions, liquidity constraints, and the distribution of money holdings jointly shape consumption and welfare. Early search-theoretic contributions generate a non-degenerate distribution of money by combining idiosyncratic trading opportunities with incomplete insurance, and develop computational methods to characterize stationary equilibria and policy effects in economies with cross-sectional heterogeneity (e.g., [Molico, 2006](#); [Chiu and Molico, 2010, 2011](#); [Menzio, Shi and Sun, 2013](#); [Bethune and Rocheteau, 2023](#)). A related line emphasizes the distributional consequences of inflation and monetary transfers when liquidity is unevenly held across households (e.g., [Erosa and Ventura, 2002](#); [Camera and Chen, 2014](#)). Across these papers, a central message is that once the distribution of liquid assets is non-degenerate, monetary policy has inherently redistributive effects and welfare cannot be understood from representative-agent logic alone.

Our framework is closely related to [Rocheteau et al. \(2021\)](#), which provides a tractable heterogeneous-agent New Monetarist environment in which households face idiosyncratic spending opportunities and cannot fully neutralize liquidity shocks through labor adjustment, leading to a stationary distribution of money holdings. Their analysis asks how monetary policy affects the economy when the distribution of liquidity is an endogenous state variable, and highlights the importance of tracking distribution dynamics for both short-run and long-run policy effects. We adopt the same broad strategy to generate heterogeneity—idiosyncratic trading needs combined with fixed labor income—so households differ endogenously in their liquid-wealth positions and the marginal value of liquidity varies across the cross-section. This environment is well suited for studying how monetary policy interacts with liquidity constraints, because it delivers a non-degenerate distribution of real balances in stationary equilibrium.

On the other hand, our contribution departs from [Rocheteau et al. \(2021\)](#) along two dimensions that matter for both empirical realism and for the economic question at hand. First, we assume strictly concave utility over centralized consumption rather than the linear specification often used for tractability. This preference structure strengthens the incentive of liquidity-poor households to run down balances, supports an endogenous hand-to-mouth mass, and helps generate a right-skewed liquid-asset distribution that is closer to the data. Second, the central object of our analysis is competitive banking and its liquidity-transformation role. While [Rocheteau et al. \(2021\)](#) focuses on how labor-market frictions and the resulting incomplete insurance shape monetary-policy transmission, we embed a competitive banking sector that offers deposits and credit and study how financial intermediation reshapes both the use and the distribution of liquidity. This shift in focus is crucial: banking changes equilibrium money demand and the inflation-tax base, and it generates welfare effects that reflect the interaction of state-contingent insurance with endogenous distributional reweighting—channels that are absent in representative-agent banking models and not the primary focus of RWW.

The existing literature has made significant progress in incorporating heterogeneity into monetary models to study the distributional effects of inflation. For example, [Erosa and Ventura \(2002\)](#) interpret inflation as a regressive consumption tax, emphasizing how the burden varies systematically across wealth levels. In search-theoretic settings, [Molico \(2006\)](#) and [Chiu and Molico \(2010\)](#) develop frameworks with endogenous money distributions and show how liquidity constraints shape

welfare and, in some environments, generate price dispersion. Camera and Chen (2014) extend this line by analyzing the long-run heterogeneous effects of inflation on wealth and consumption inequality. More recently, studies such as Cao, Meh, Ríos-Rull and Terajima (2021) and Bethune and Rocheteau (2023) examine how financial innovations and asset liquidity interact with inflation to alter welfare and redistribution. Our paper contributes to this literature by studying a specific and empirically relevant redistribution channel that has received less attention in heterogeneous monetary models: competitive banking as a liquidity-transformation technology. By allowing households to borrow when liquidity needs are high and to earn returns on idle balances when liquidity needs are low, competitive banking changes the cross-sectional incidence of the inflation tax and the associated transfer rebate, and thus reshapes who gains and who loses from inflation. Importantly, once the liquidity distribution is endogenous, the welfare implications of banking are not uniform across policy regimes: the relative importance of deposit-based versus credit-based gains varies systematically with inflation, generating an inflation-dependent composition of welfare effects that is absent from representative-agent banking analyses.

The rest of the paper is organized as follows. Section 2 describes the model and the stationary monetary equilibrium. Section 3 presents the result of the quantitative analysis. Section 4 concludes the paper.

2 Model

Our framework builds on the monetary model of Berentsen et al. (2007) and extends it to a heterogeneous-agent environment in the spirit of Rocheteau et al. (2021). The key purpose of introducing heterogeneity is to study the role of financial intermediation in shaping the distribution of liquid assets, consumption behavior, and welfare. As in BCW, trade takes place in a decentralized market (DM) and a centralized market (CM), and competitive banks provide both deposit and lending services. Unlike the representative-agent setting in BCW, however, households in our economy differ in their money holdings and access to credit, which allows the model to generate hand-to-mouth borrowers and depositors within a unified framework. Throughout, we work directly with real variables, expressing all nominal quantities in terms of real balances.

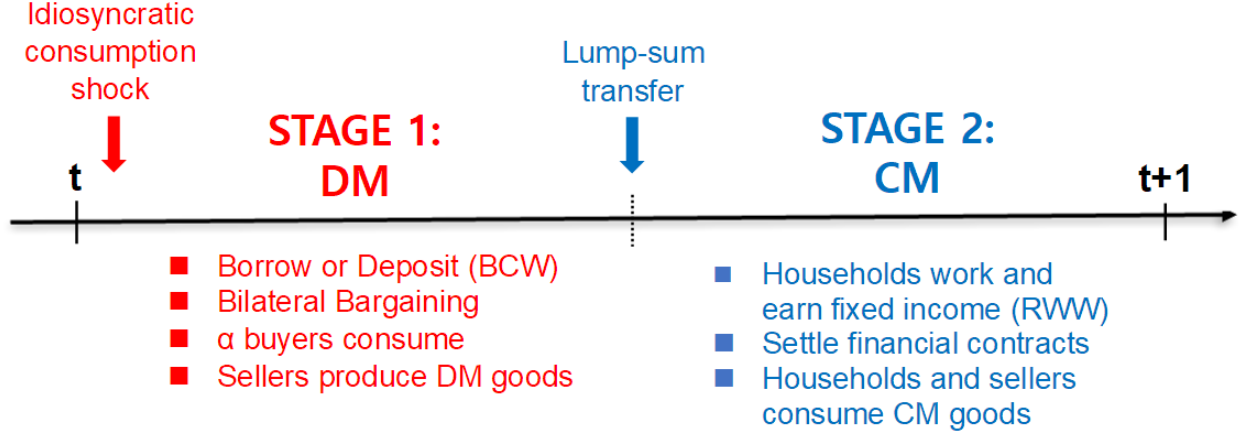
In Section 2.1, we introduce the model environment, and in Section 2.2, we present the optimization problem of the household. Section 2.3 discusses the bank’s problem, while Section 2.4 introduces the distribution of real balances. Section 2.5 presents the market clearing conditions and defines stationary monetary equilibrium.

2.1 Environment

Timing

The economy in this model is made up of two groups: households and sellers. Time is divided into discrete periods, each comprising a decentralized market (DM) and a centralized market (CM). Both DM and CM goods are perishable. The use of money as a medium of exchange in the DM is necessitated by informational frictions, such as lack of commitment or double coincidence problem.

Figure 2: Timing



Meanwhile, banks in this model are capable of keeping track of agents' financial records following [Berentsen et al. \(2007\)](#). These distinct informational assumptions on the goods markets and banking sector allow for the simultaneous existence of money and credit in the model ([Berentsen et al. \(2007\)](#)). The discount factor across periods is β , which is between 0 and 1.

The timing in the model is detailed in Figure 2. At the start of each period's DM, households face idiosyncratic consumption shocks which are random and independent across households and periods. Some households (those with probability α) receive the opportunity to consume, while others with probability $1 - \alpha$ cannot. Following the realization of the consumption shock, households decide whether to borrow from or save in the bank. Subsequently, buyers and sellers are randomly matched and engage in bilateral bargaining.

After the DM concludes, households and sellers enter the CM. In the CM, households receive lump-sum transfers from the monetary authority and earn fixed labor income $\bar{h}$. The fixed labor income implies that households cannot fully undo their DM consumption shocks by earning as much as they desire. Households use one unit of labor to produce one unit of CM goods. Households also settle any financial contracts made during the DM with the banks: borrowers pay back loans with interest, while depositors receive returns on their savings. Finally, households and sellers consume CM goods and carry over any remaining real balances to the next period.

Preferences

In this model, households derive utility from both DM and CM consumption. The utility from DM consumption, represented by $v(y)$, and the utility from CM consumption, represented by $u(c)$, are both continuously differentiable, strictly increasing, $v', u' > 0$, and concave, $v'', u'' < 0$ with boundary conditions $v'(+\infty) = u'(+\infty) = 0$ and $v'(0) = u'(0) = +\infty$. For the analysis below, we maintain these general assumptions and, in particular, keep CM utility strictly concave. This contrasts with [Rocheteau et al. \(2021\)](#), who impose a linear CM utility, $u(c) = c$, for analytical tractability. The strict concavity of $u(\cdot)$ in our environment implies that households optimally run down liquid balances through CM consumption, which contributes to a right-skewed distribution of

money holdings, whereas the linear specification in Rocheteau et al. (2021) generates a left-skewed distribution in their framework.

In the CM, households supply a fixed amount of labor and produce a fixed output, represented by $\bar{h}$. One unit of labor produces one unit of CM goods and incurs one unit of disutility. Thus, the per-period utility of a household is

$$v(y) + u(c) - \bar{h}. \quad (2.1)$$

On the other hand, sellers incur a cost of producing DM goods, denoted by $q(y)$. They are assumed to be risk-neutral, deriving utility from CM consumption equal to the quantity consumed. Thus, the per-period utility of a seller is

$$-q(y) + c. \quad (2.2)$$

Sellers spend all the money earned from DM trades on CM goods, as they have no incentive to carry over real balances to the next period due to the effects of inflation and time discounting. Hence, the analysis primarily focuses on households, as the seller's problem is relatively straightforward.

Monetary policy

In this economy, monetary policy is conducted by adjusting the growth rate of the money supply. The total money stock, denoted by M , grows at a constant rate τ , according to the process:

$$\frac{M_{+1}}{M} = 1 + \tau \quad (2.3)$$

At the beginning of each period's CM, households receive a lump-sum monetary transfer equal to τM . We focus on the long-run stationary equilibrium in which the total value of the money stock remains constant over time, represented by $\phi M = \phi_{+1} M_{+1}$, where ϕ is the value of money. The gross rate of return on money is given by $R = \phi_{+1}/\phi$. Since the total value of money is constant over time, the value of the monetary transfer, $\phi \tau M$, is also constant in each period. In this equilibrium, the increase in the price level, represented by ϕ/ϕ_{+1} is equal to the growth rate of money stock, M_{+1}/M . As a result, the inflation rate equals τ , as determined by the monetary policy rule.

2.2 Household

We describe the household's optimization problem using backward induction, starting with the centralized market (CM). As will be elaborated later, our model defines hand-to-mouth households as those who choose zero next-period real balances ($z' = 0$) in the CM.¹ *Hand-to-mouth* households

¹ In our model, hand-to-mouth households are defined based solely on liquid assets, unlike Kaplan, Violante and Weidner (2014) and HANK literature, who consider both liquid and non-liquid assets; this focus is appropriate as it aligns with the model's objective to analyze liquidity constraints and immediate consumption behavior.

enter the subsequent DM with no real balances, effectively holding zero liquid assets.

2.2.1 Household in CM

Households choose consumption and money holdings in the CM. Each household is characterized by their real money balance z , loan l , and deposit d in the DM. The value function at the start of the CM is represented by $W(z, l, d)$ and is determined as follows:

$$\begin{aligned} W(z, l, d) &= \max_{c, z' \geq 0} \{u(c) - \bar{h} + \beta V(z')\} \\ \text{s.t. } z' &= R(z + \bar{h} + T + (1 + i_d)d - (1 + i)l - c) \geq 0 \end{aligned} \quad (2.4)$$

where V is the DM value function, z' is the next period real balances, i_d and i are the deposit and loan interest rate and T is the real value of monetary transfer. The variables are all in real terms. R is exogenously determined by the monetary authority. i_d and i are endogenously determined from loan market clearing condition. Given prices of i_d, i , and R , the households maximize the utility from CM consumption $u(c)$ minus the disutility of labor $\bar{h}$ and the discounted continuation value.

After introducing Lagrange multipliers for the budget constraint and formulating the Lagrangian, we derive the envelope conditions by differentiating with respect to the z, l, d , and c . The envelope conditions are as follows:

$$W_z = u'(c) \quad (2.5)$$

$$W_d = u'(c) \cdot (1 + i_d) \quad (2.6)$$

$$W_l = -u'(c) \cdot (1 + i) \quad (2.7)$$

Unlike [Lagos and Wright \(2005\)](#) where the value function has a linear form with $W_z = 1$ due to quasi-linear preferences and flexible labor supply, the envelope conditions in our model show that $W_z = u'(c)$ as in Equation (2.5). This crucial difference arises since the model assumes fixed labor supply ($\bar{h}$), which prevents agents from adjusting their labor hours to absorb wealth shocks. In the standard [Lagos and Wright \(2005\)](#) model, the combination of quasi-linear utility and flexible labor allows agents to optimally adjust their labor supply, eliminating wealth effects, thereby leading to a degenerate distribution of money holdings. On the other hand, with fixed labor supply in our model, agents cannot use labor adjustment as a buffer against wealth shocks, resulting in a non-degenerate distribution where the marginal value of wealth equals the marginal utility of consumption. This feature is essential for generating *hand-to-mouth* agents, which are key empirical patterns that the standard [Lagos and Wright \(2005\)](#) model cannot capture.

Household's Optimality in CM

We will now examine the optimality conditions in the CM, with a particular focus on analyzing the optimal behavior of *hand-to-mouth* and *non-hand-to-mouth* households, which is the primary objective of our paper. Substituting the budget constraint into the objective function,

$$W(z, l, d) = \max_{c, z'} \{u(c) - \bar{h} + \beta V(z') + \xi [z + \bar{h} + T + (1 + i_d)d - (1 + i)l - c - R^{-1}z']\} \quad (2.8)$$

where ξ denotes the Lagrange multiplier associated with the condition $z' \geq 0$.

In our model, households are classified as either *hand-to-mouth* or *non-hand-to-mouth* household based on liquidity constraints, driven by fixed labor income and the availability of credit as follows.

First, when the constraint is not binding ($\xi = 0$), the optimal condition for non-hand-to-mouth households is derived as follows.

$$\begin{aligned} R\beta V'(z') &= u'(c) \\ &= u'(z + \bar{h} + T + (1 + i_d)d - (1 + i)l - R^{-1}z') \end{aligned} \quad (2.9)$$

LHS of Equation (2.9) is the marginal benefit of holding real balances and the RHS can be interpreted as the marginal cost of holding real balances. This condition (2.9) implies that the marginal benefit of holding real balances must be equal to the marginal cost of holding real balances. The agents hold heterogeneous amounts of money at the end of the period since each agent receives the idiosyncratic consumption shock and fixed labor income. Under this interior solution, households are classified as *non-hand-to-mouth*, as they choose to save a strictly positive amount of liquid assets ($z' > 0$).

Next, let us examine the case of hand-to-mouth households. If the constraint is binding (i.e., $\xi > 0$), the households' optimal consumption (c) and next period real balances (z') are determined by the following conditions.

$$R\beta V'(0) < u'(z + \bar{h} + T + (1 + i_d)d - (1 + i)l) \quad (2.10)$$

$$z' = 0 \quad (2.11)$$

$$c = z + \bar{h} + T + (1 + i_d)d - (1 + i)l \quad (2.12)$$

These households, referred to as *hand-to-mouth* agents, consume all their disposable resources after financial settlements in the CM, as the marginal utility of current consumption exceeds the discounted marginal benefit of carrying money into the next period. *Hand-to-mouth* households optimally choose to carry zero liquidity forward due to the availability of credit and the fixed labor income. Equation (2.10) implies that the marginal utility of depleting the last unit of money exceeds the marginal cost of doing so. From the condition (2.10) and the assumption of competitive banks ($i = i_d$), the following condition must be satisfied for households to be *hand-to-mouth* household in the CM:

$$d - l < \frac{1}{1 + i} (u'^{-1}(R\beta V'(0)) - z - \bar{h} - T) \quad (2.13)$$

Equation (2.13) implies net deposit $(d - l)$ is smaller than a certain threshold for households to be *hand-to-mouth* in the CM. If the current liquidity holdings (z) are low or wage income $(\bar{h})$ is insufficient, the household is more likely to become a *hand-to-mouth* household. Similarly, if deposits (d) are low or loans (l) are substantial, the household is also more likely to be *hand-to-mouth*.

We derive the cut-off for net deposit which distinguishes *non-hand-to-mouth* and *hand-to-mouth* households as follows:²

$$\tilde{l} = \frac{1}{1+i} (u'^{-1}(R\beta V'(0)) - \bar{h} - T) \quad (2.14)$$

To summarize, the decision rules for *non-hand-to-mouth* households and *hand-to-mouth* households are as follows:

Lemma 1. *With $\tilde{l} = \frac{1}{1+i} (u'^{-1}(R\beta V'(0)) - \bar{h} - T)$, decision rules in CM are as follows:*

1. (**Non Hand-to-Mouth Household**) *If $\tilde{l} < d - l$, then the CM decision rule follows $R\beta V'(z') = u'(c)$ where $z' > 0$.*
2. (**Hand-to-Mouth Household**) *If $d - l \leq \tilde{l}$, then the CM decision rule follows $z' = 0$ and $c = \bar{h} + T + (1+i)(d-l)$.*

Figure 3 illustrates the household decision rule, derived from Lemma 1's characterization of optimal policy functions.³ The left panel in Figure 3 illustrates the CM consumption policy as a function of net deposits, while the right panel characterizes the corresponding optimal choice of next-period real balances. When a net deposit exceeds the threshold $\tilde{l}$, households exhibit standard consumption smoothing behavior with positive real money balances ($z' > 0$) consistent with the Euler equation $R\beta V'(z') = u'(c)$. On the other hand, households with net deposits below $\tilde{l}$ become *hand-to-mouth*, consuming their entire resources (i.e., $c = \bar{h} + T + (1+i)(d-l)$) and carrying zero money balances forward ($z' = 0$).

The CM policy functions reveal critical differences in consumption and saving behaviors across *non-hand-to-mouth* and *hand-to-mouth* households. For *non-hand-to-mouth* household, consumption decisions align the marginal utility of consumption with the discounted marginal benefit of savings. This results in a smooth consumption function characterized by efficient resource allocation. On the other hand, for *hand-to-mouth* households ($d - l \leq \tilde{l}$), the inability to save or smooth consumption compels them to consume their entire disposable income and liquid assets, producing a steeper consumption with a slope of $1 + i$.

2.2.2 Household in DM

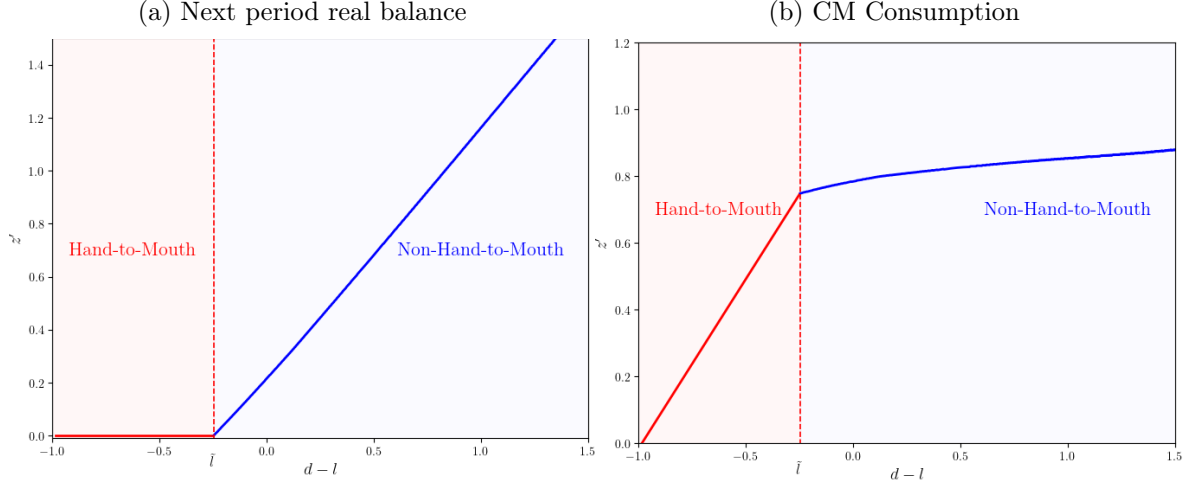
Terms of Trade

In the DM, transactions take place through bilateral matches between buyers and sellers. The terms of trades in pairwise meetings are set by buyers. Each matched buyer makes a take-it-or-

²With the presence of banking institutions, all agents optimally choose to carry zero real balances ($z = 0$) into the CM. This follows from the dominance of saving through banks over holding money, as the return on deposits strictly dominates the return on money holdings whenever nominal interest rates are strictly positive.

³These policy functions are derived under the calibrated parameter values specified in Section A.3

Figure 3: CM policy functions



leave-it offer, (y, p) , to a seller, where y is the DM output to be produced by the seller in exchange for payment, p , in the form of real balances. In this case, the seller's participation constraint is $-q(y) + p \geq 0$ and it holds at equality since the buyer has all the bargaining power. For simplification, if we assume a one-to-one relationship between the production of DM output and the disutility, $q(y) = y$, then $p = y$ holds.

DM life time utility

The lifetime expected utility of a household who has z real balance at the beginning of the period is

$$V(z) = \alpha \cdot \max_{y,l,d} \{v(y) + W(z + l - d - y, l, d)\} + (1 - \alpha) \cdot \max_{\tilde{d}} W(z - \tilde{d}, 0, \tilde{d}) \quad (2.15)$$

The first term on the RHS is the weighted lifetime utility of a buyer and the second term is the weighted lifetime utility of a non-buyer. A non-buyer deposits all the money in the bank to earn deposit gains as long as the interest rate is positive. If the interest rate is zero, depositors are indifferent between depositing in the bank or holding real balances.

A non-buyer does not trade in the decentralized market and therefore has no motive to retain liquidity for within-period purchases. With probability $1 - \alpha$, the household is inactive and chooses how much of its beginning-of-period real balances z to deposit in the bank, denoted $\tilde{d} \in [0, z]$, before entering the centralized market with portfolio $(z - \tilde{d}, 0, \tilde{d})$. Since idle money is eroded by the inflation tax while deposits earn the gross return $1 + i_d$, it is optimal for an inactive household to deposit all available balances whenever $i_d > 0$. Hence, for $i_d > 0$ the optimal policy is $\tilde{d}(z) = z$, so the household enters the centralized market holding deposits rather than cash; when $i_d = 0$, it is indifferent between depositing and carrying money forward.

The problem for buyer is to choose DM consumption, y , DM loan, l , and DM deposit, d , to maximize their lifetime utility,

$$\max_{y,l,d} v(y) + W(z + l - d - y, l, d) \quad (2.16)$$

$$s.t. \quad y \leq z + l - d \quad (2.17)$$

$$0 \leq d \leq z \quad (2.18)$$

$$0 \leq l \leq \bar{l} \quad (2.19)$$

$$\bar{l} = \psi \cdot \frac{\bar{h}+T}{1+i}, \quad \text{where } \psi \in [0, 1]. \quad (2.20)$$

Equation (2.17) implies that it is always optimal for the household to fully deplete all the money as long as the interest rate is positive. If the partial depletion is optimal choice, it means the former decision of borrowing or deposit was not optimal. The constraint (2.17) implies that the deposit must be less than the money holding is not binding as it is always optimal for households to consume in the DM. There is a borrowing constraint, $l \leq \bar{l}$, which is associated with the income, $\bar{h}$, plus transfer, T . If buyers borrow more than $\frac{\bar{h}+T}{1+i}$, they cannot fully pay back the loan in the CM. ψ is the exogenously imposed ratio that buyers can borrow out of maximum amount of loan. If $\psi = 1$, buyers can borrow up to the maximum amount that they can possibly pay back. We assume $\psi = 1$ in the baseline analysis. If $\psi = 0$, then buyers cannot borrow at all. It is the case where there is no bank in the economy. We compare the no-bank economy ($\psi = 0$) with the baseline economy ($\psi = 1$) to assess the role of banking in the model.

To derive the buyer's optimal decision rule, we formulate the Lagrangian as follows

$$L = v(y) + W(z + l - d - y, l, d) + \lambda [z + l - d - y] + \lambda_l [\bar{l} - l] + \lambda_d d + \lambda_0 l \quad (2.21)$$

where λ represents the Lagrangian multiplier associated with the liquidity constraint, λ_l and λ_0 correspond to the borrowing constraints, and λ_d denotes the multiplier for the deposit constraint. Taking the derivative of Equation (2.21) with respect to z, l , and d and employing the envelope conditions from Equations (2.5), (2.6), and (2.7), we obtain following optimal decision condition

$$(1 + i)u'(c) = v'(y) - \lambda_l \quad (2.22)$$

LHS of Equation (2.22) is the marginal cost of borrowing one unit of real balance while the RHS is the marginal utility of borrowing one unit of real balance. Note that $1 + i$ can be interpreted as the relative price of DM goods to CM goods.

First, when the borrowing constraint is binding (i.e., $\lambda_l > 0$), the optimality condition is as follows:

$$(1 + i)u'(c) + \lambda_l = v'(y) \quad (2.23)$$

Equation (2.23) implies households want to borrow more, but cannot because of the binding borrowing constraint. In this case, the marginal utility is larger than the marginal cost, such that $v'(y) > (1 + i)u'(c)$. When the borrowing constraint is binding, constrained agents are not able

to borrow until the marginal utility goes down to the level of the marginal cost of borrowing. However, this case cannot be supported in this model due to the boundary condition imposed on the CM utility. Households refrain from borrowing up to the limit to avoid zero consumption in the CM.

Second, when the borrowing constraint is non-binding (i.e., $\lambda_l = 0$), we obtain the following optimality condition:

$$(1 + i)u'(c) = v'(y) \quad (2.24)$$

Combining with Equation (2.24) and the CM optimization conditions from Equations (2.9) and (2.10),

$$(1 + i)R\beta V'(z') \leq v'(y) \quad " = " \text{ if } z' > 0 \quad (2.25)$$

This implies that DM consumption function of liquidity-constrained households in the CM is differentiated from that of unconstrained households.

From the Equation (2.24) and (2.12), a constrained household's choice of DM consumption satisfies

$$y(z) = v'^{-1}((1 + i)u'(z + \bar{h} + T - (1 + i)l)) \quad (2.26)$$

In the DM, agents are classified into three distinct categories based on their real balance holdings z : (potentially) hand-to-mouth borrowers, (non-hand-to-mouth) borrowers, and depositors. *Hand-to-mouth borrowers* represent a particularly interesting case where agents who borrow in the DM subsequently become hand-to-mouth in the CM. More precisely, these agents enter the DM with insufficient liquidity, requiring them to borrow, which in turn leads to lower net deposits when they enter the subsequent CM.

Using Equation (2.25) and the hand-to-mouth borrowing condition from Equation (2.14), along with the constraint $z' = 0$, we identify a cutoff level of real balances, denoted by $\bar{z}$, below which households are considered hand-to-mouth, as they are going to choose not to carry positive real balances into the next period.

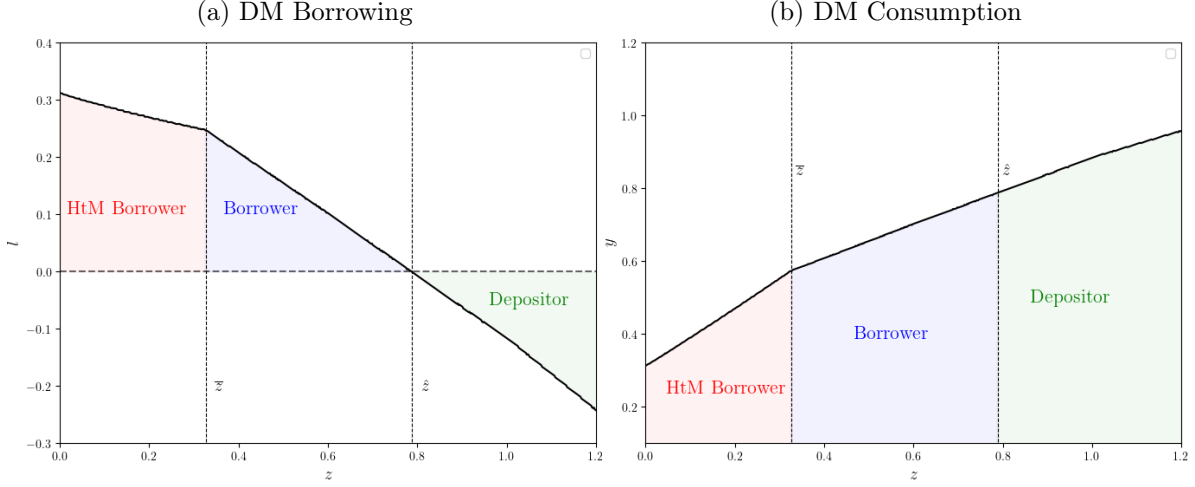
$$\bar{z} = v'^{-1}((1 + i)R\beta V'(0)) - \tilde{l} \quad (2.27)$$

Next, when a buyer has $\hat{z}$ which is the solution to $v'(\hat{z}) = W_d(0, 0, 0)$ ($:= u'(\bar{h} + T) \cdot (1 + i_d)$), then he neither borrows nor deposits but consumes all his money in the DM. When a buyer holds z such that $v'(z) > W_d(0, 0, 0)$, he borrows up to the point where the marginal utility goes down to the level of marginal cost of consumption, i.e., $v'(\hat{z} + l) = W_d(0, l, 0)$. When $v'(z) < W_d(0, 0, 0)$, then he deposits until marginal utility and marginal cost are equal, i.e., $v'(\hat{z} - d) = W_d(0, 0, d)$. We summarize the optimal DM decision rules in the Lemma 2.

Lemma 2. *With $\bar{z} = v'^{-1}((1 + i)R\beta V'(0)) - \tilde{l}$ and $\hat{z} = v'^{-1}(W_d(0, 0, 0))$, buyer's consumption in DM is derived as follows*

1. **(Hand-to-Mouth Borrower)** For $0 \leq z < \bar{z}$, $y(z) = z + \tilde{l}$ such that Equation (2.22)

Figure 4: DM policy functions



holds.

2. **(Borrower)** For $\bar{z} \leq z < \hat{z}$, $y(z) = z + l(z)$ such that Equation (2.24) holds.
3. **(Depositor)** For $\hat{z} \leq z$, $y(z) = z - d(z)$ such that Equation (2.24) holds

Figure 4 presents the policy functions in the DM, with the left panel depicting DM borrowing behavior as a function of real balances and the right panel illustrating the corresponding DM consumption patterns.⁴ Figure 4 highlights three distinct regions: the blue area represents hand-to-mouth borrowers, the yellow area indicates non-hand-to-mouth borrowers, and the green area denotes depositors. These classifications correspond to theoretical characterization established in the Lemma 2. For low real balances ($0 \leq z < \bar{z}$), hand-to-mouth borrowers exhibit a borrowing function that fully depletes their future money holdings, where consumption equals initial real balances plus borrowed amount ($y(z) = z + \tilde{l}$). In the intermediate region ($\bar{z} \leq z < \hat{z}$), borrowers optimize according to the standard Euler equation while maintaining positive future money holdings. Finally, for sufficiently high real balances ($\hat{z} \leq z$), agents become depositors, with consumption determined by the optimal deposit choice ($y(z) = z - d(z)$).

2.3 Bank's problem

Following the framework of Berentsen et al. (2007), we assume a perfectly competitive banking sector that accepts deposits and issues loans. The representative bank solves the following problem:

$$\begin{aligned} \max_{L,D} \quad & iL - i_d D \\ \text{s.t.} \quad & L \leq D \end{aligned} \tag{2.28}$$

where D is total deposits and L is total loans. The banks choose the amount of loans and deposits to maximize profits. However, due to perfect competition, banks earn zero profit in equilibrium, implying that the borrowing rate equals the deposit rate. Therefore, the condition $i = i_d$ holds, and

⁴These policy functions are derived under the calibrated parameter values specified in Section A.3

the equilibrium interest rate is determined by the market-clearing condition in the loan market. The model also allows for the possibility of a zero lower bound (ZLB) on the interest rate, in which case the loan market may fail to clear even when the interest rate reaches zero.

2.4 Distribution of real balances

$G(x)$ is the measure of households holding no more than x net deposit at the beginning of the CM stage. Then it solves:

$$G(x) = \int [\alpha \mathbb{I}_{\{d(z)-l(z) \leq x\}} + (1-\alpha) \mathbb{I}_{\{z \leq x\}}] dF(z) \quad (2.29)$$

where $F(z)$ is the cumulative money holding distribution at the beginning of the DM. The first term on the right side of Equation (2.29) indicates the measure of households who receive the consumption shock in the DM. These agents enter CM with net deposit $d(z) - l(z)$ where z is the real balance at the start of the DM. On the other hand, the second term on the right side of Equation (2.29) captures the measure of households that do not get the opportunity to trade in the DM. In this case, the households enter the CM with z net deposit. The distribution of real balances at the start of the DM is given by:

$$F_+(z) = \int \mathbb{I}_{\{z'(x) \leq z\}} dG(x) \quad (2.30)$$

where $z'(x)$ is the CM choice of real balances given x net deposit. The distributions, $F(z)$ and $G(x)$, should follow the law of motion, Equation (2.29) and (2.30), in each period. The probability density functions (PDFs) for these cases are characterized as follows:

$$f(z) \left(= \frac{dF(z)}{dz} \right) = \frac{\partial F(z)}{\partial z} \quad (2.31)$$

$$g(x) \left(= \frac{dG(x)}{dx} \right) = \frac{\partial G(x)}{\partial x} \quad (2.32)$$

2.5 Stationary Monetary Equilibrium

The CM market clearing condition states that the total consumption of CM goods by households and sellers must equal the total amount of CM goods produced in the economy. This is represented by the equation:

$$\int c(x) dG(x) + \alpha \int y(z) dF(z) = \bar{h} \quad (2.33)$$

On the LHS of the equation (2.33), the first term represents the consumption of goods by households in the CM, while the second term represents the consumption of goods by sellers who have traded in the DM and now spend their money in the CM. The parameter α measures the proportion

of sellers who engage in DM transactions. The right-hand side of the equation represents the total amount of goods produced in the CM. This equation ensures that all goods produced are consumed, and there is no excess or shortage in the market.

The money market clearing condition states that the total real value of the money supply, represented by Z , must be equal to the total real demand for money, represented by the integral of the real money holding distribution, $F(z)$, over all households.

$$Z = \int z dF(z) \quad (2.34)$$

This ensures that the money market is in equilibrium and that the supply of money is equal to the demand for money in the economy.

The loan market clearing condition can be characterized into two types of equilibria, one where the interest rate is strictly positive and the loan market is cleared, and another where the interest rate is at the zero lower bound and there is excess deposit in the economy. The first type of equilibrium is as follows.

$$i > 0 \quad (2.35)$$

$$\alpha \int (d(z) - l(z)) dF(z) + (1 - \alpha) \int \tilde{d}(z) dF(z) = 0 \quad (2.36)$$

This case is when strictly positive interest rate clears the loan market. The LHS of (2.36) is basically net deposit of the economy where the first term captures the net deposit of buyers and the second term indicates the deposit of non-buyers. The second type of equilibrium is as follows.

$$i = 0 \quad (2.37)$$

$$\alpha \int d(z) - l(z) dF(z) + (1 - \alpha) \int \tilde{d}(z) dF(z) > 0 \quad (2.38)$$

In this case, the equilibrium interest rate is equal to zero and there is excess deposit in the economy. This equilibrium can be supported because of the assumption that people deposit rather than hold their money when the interest rate is zero. Note that the negative interest rate however cannot be supported in the equilibrium because no one is going to deposit their money when interest rate is negative and the bank cannot exist accordingly in the economy.

We now describe a formal definition of the Stationary Monetary Equilibrium (SME) that characterizes the steady-state allocation.

Definition 1. *Given the exogenous monetary policy τ , the Stationary Monetary Equilibrium (SME) is characterized by a tuple $V, W, y, l, d, c, z', R, i, F, G$ consisting of:*

1. *Value functions $\{V, W\}$ that satisfy the recursive Bellman equations specified in Equations (2.8) and (2.15)*
2. *DM policy functions $\{y, l, d\}$ solving the optimal portfolio and consumption decisions described in Lemma 2*
3. *The CM policy functions $\{c, z'\}$ satisfying the optimal allocation detailed in Lemma 1*
4. *The rate of return of money $\{R\}$ that clears the money market according to Equation (2.34)*
5. *The interest rate $\{i\}$ that solves the loan market-clearing condition as in Equation (2.36)*

6. *Distribution functions $\{F, G\}$ satisfying the evolution of real money balances according to the law of motion in Equation (2.30).*

3 Welfare Implication of Banking in a Heterogeneous-Agent Economy

In this section, we quantitatively assess the intermediation role of competitive banking in our heterogeneous-agent model. Our main objective is to clarify how the welfare and distributional implications of banking differ from the representative-agent benchmark in ‘Berentsen et al. (2007) (BCW).

Our quantitative analysis is based on a U.S. calibration; full details of the calibration and parameter choices are provided in the Appendix A.3. The section is organized as follows. Section 3.1 studies how competitive banking changes money holdings and the stationary distribution of real balances. Section 3.2 then explains the mechanisms behind these patterns, emphasizing two forces—an insurance effect and a distributional effect. Finally, Section 3.3 examines the welfare implications of banking using our welfare measures and decompositions.

Before presenting the main results, we first provide an overview of the self-insurance (pure-currency) economy that serves as our benchmark without banking.

Self-insurance (i.e., pure currency) economy

Before turning to these results, we briefly describe the self-insurance benchmark economy without banking. This pure-currency economy is closely related to Rocheteau et al. (2021). We begin by specifying the CM value function $\tilde{W}(z)$ for self-insuring households.

$$\begin{aligned} \tilde{W}(z) &= \max_{\tilde{c}, \tilde{z}' \geq 0} \left\{ u(\tilde{c}) - \bar{h} + \beta \tilde{V}(\tilde{z}') \right\} \\ \text{s.t. } \tilde{z}' &= R(z + \bar{h} + T - \tilde{c}) \geq 0 \end{aligned} \quad (3.1)$$

Subsequently, we characterize the DM value function $\tilde{V}$ through the following recursive formulation.

$$\tilde{V}(z) = \alpha \cdot \max_{\hat{y}} \left\{ v(\hat{y}) + \tilde{W}(z - \hat{y}) \right\} + (1 - \alpha) \cdot \tilde{W}(z) \quad (3.2)$$

Differentiating Equation (3.2) with respect to z yields the following expression for the marginal value of money as follows:

$$\tilde{V}'(z) = \alpha v'(z) + (1 - \alpha) \tilde{W}'(z) \quad (3.3)$$

The optimization problem for buyer is to choose DM consumption, $\hat{y}$, to maximize their lifetime

utility,

$$\begin{aligned} \max_{\hat{y}} \quad & v(\hat{y}) + \tilde{W}(z - \hat{y}) \\ \text{s.t.} \quad & \hat{y} \leq z \end{aligned} \tag{3.4}$$

$\tilde{G}(\tilde{x})$ is the measure of households holding no more than $\tilde{x}$ real balance at the beginning of the CM stage. Then it solves:

$$\tilde{G}(\tilde{x}) = \int [\alpha \mathbb{I}_{\{z - \hat{y} \leq \tilde{x}\}} + (1 - \alpha) \mathbb{I}_{\{z \leq \tilde{x}\}}] d\tilde{F}(z) \tag{3.5}$$

where $\tilde{F}(z)$ is the cumulative money holding distribution at the beginning of the DM. The distribution of real balances at the start of the DM is given by:

$$\tilde{F}_+(z) = \int \mathbb{I}_{\{z'(\tilde{x}) \leq z\}} d\tilde{G}(\tilde{x}) \tag{3.6}$$

3.1 Banking, Money Demand, and the Distribution of Money Holdings

This section shows that, in our heterogeneous-agent economy, bank credit substitutes for cash self-insurance, lowering average real balances in equilibrium. This result contrasts with the representative-agent logic in [Berentsen et al. \(2007\)](#), where deposit returns strengthen incentives to carry liquidity across periods and thereby raise money demand. In our environment, deposit services are still present, but the dominant quantitative force operates at the lower tail: access to credit generates a positive mass of hand-to-mouth households who optimally hold zero real balances, shifting the stationary distribution toward low balances and reducing average money holdings.

Banking introduces two opposing incentives for holding real balances. On the one hand, deposit services raise the return to carrying liquidity into periods in which households do not trade in the DM. This strengthens the incentive to retain liquidity across periods, pushing money holdings upward for households with sufficiently high balances. On the other hand, access to credit relaxes the need for precautionary cash holdings when a DM trading opportunity arrives. Liquidity-poor households can finance DM purchases by borrowing, which makes it optimal for some households to carry little or even zero money into the next period. Our goal in this subsection is to make this trade-off by comparing the marginal value of real balances in the banking economy and in the self-insurance (pure-currency) benchmark.

To formalize the mechanism, we characterize the marginal value of real balances at the beginning of the DM. Differentiating the household lifetime value in equation (2.15) with respect to z yields

$$V'(z) = \underbrace{\alpha v'(y(z))}_{|A|: \text{ DM-trade state}} + \underbrace{(1 - \alpha)(1 + i)u'(\tilde{c}(z))}_{|B|: \text{ non-trade (deposit) state}}. \quad (3.7)$$

The term $|A|$ captures the marginal value of an additional unit of liquidity in the state where the household trades in the DM (probability α). The term $|B|$ captures the marginal value in the state where the household does not trade (probability $1 - \alpha$) and can deposit its liquidity to earn the gross return $(1 + i)$ before consuming in the CM. Holding fixed CM marginal utility, the deposit return scales up $|B|$, reflecting the BCW-style force that makes carrying liquidity forward more attractive.

Access to credit lets some households hold zero money, generating endogenous hand-to-mouth mass at the lower tail and compressing equilibrium money demand. Evaluating the marginal value at zero balances in the banking economy gives

$$V'(0) = \underbrace{\alpha v'(l)}_{|A| < +\infty} + (1 - \alpha)(1 + i)u'(\bar{h} + T - R^{-1}z') \quad (3.8)$$

Because households can borrow when a DM opportunity arrives, they can choose strictly positive DM consumption even at $z = 0$ (i.e., $l > 0$). Together with standard boundary conditions, this implies that the marginal value at the lower bound is finite, $V'(0) < +\infty$. A finite $V'(0)$ makes the corner choice of ending the CM with zero money feasible for a positive measure of households. Intuitively, when the marginal utility of spending the last unit of resources in the CM is sufficiently high relative to the discounted marginal benefit of carrying money into the next period, households optimally choose $z' = 0$ and rely on credit to finance future DM purchases. This is the hand-to-mouth margin that emerges endogenously in the banking economy.

At the same time, the deposit term $|B|$ works in the opposite direction for households with higher balances. For these households, the non-trade return $(1 + i)$ raises the payoff from carrying liquidity forward, supporting a non-trivial upper tail of money holdings. Thus, competitive banking simultaneously creates a mass of households near $z = 0$ through credit access and sustains money holdings at higher levels through deposit returns.

By contrast, in the self-insurance economy without banking, households must finance DM purchases solely with accumulated money. Evaluating the analogous marginal value at the lower bound implies

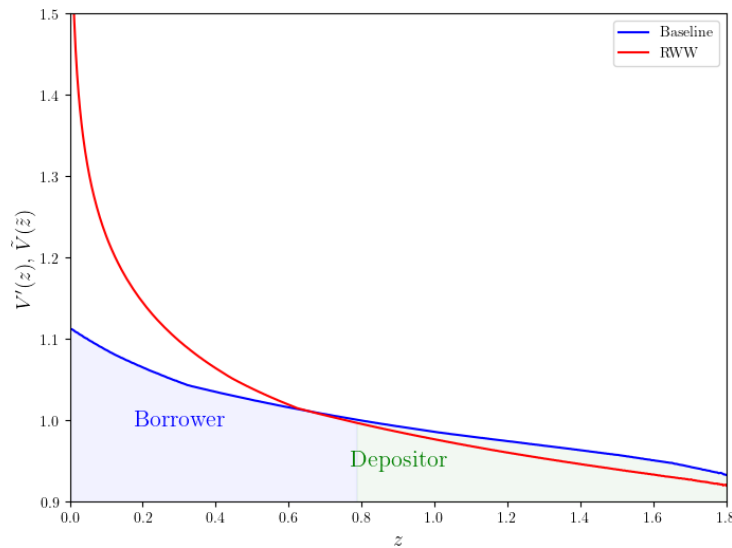
$$\tilde{V}'(0) = \underbrace{\alpha v'(0)}_{|A| = +\infty} + (1 - \alpha)\tilde{W}'(0) = +\infty. \quad (3.9)$$

Since $v'(0) = +\infty$, the marginal value of liquidity at $\tilde{z} = 0$ is infinite, which rules out hand-to-mouth behavior in stationary equilibrium: households must carry strictly positive balances to self-insure against DM trading opportunities.

Figure 5 provides a compact summary by comparing the marginal value of money across the two

economies. Figure 5 illustrates the marginal value of money as a function of real balance holdings (z) for both economies; the blue line represents the baseline economy with banking, while the red line represents the self-insurance economy. Relative to the pure-currency economy, the banking economy features a substantially lower marginal value of money at low balance levels, reflecting that credit relaxes the precautionary demand for cash and makes $V'(0)$ finite. At higher balance levels, the gap between the two marginal value functions narrows, and the deposit return term in $|B|$ supports the incentive to carry liquidity forward. The figure therefore illustrates that the trade-off that underlies our distributional results: banking weakens the marginal value of liquidity at the lower tail while preserving an incentive to hold liquidity at higher balances through deposit returns.

Figure 5: Marginal value of money



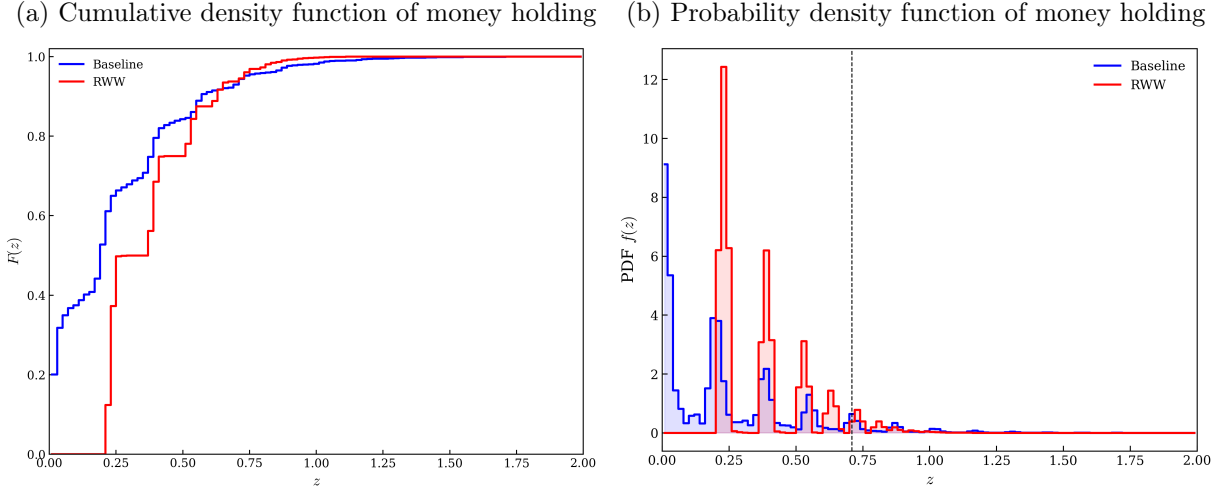
To discipline the interaction between these two countervailing banking forces and ensure the model’s empirical plausibility, we assume strictly concave CM utility ($u(c) = \log c$) unlike Rocheteau et al. (2021) who adopt linear CM utility. This preference structure is essential for generating a right-skewed stationary distribution that features a significant mass of hand-to-mouth households, as observed in the data. Under strict concavity, liquidity-poor households face high marginal utility in the CM, which sharply increases the opportunity cost of carrying money forward. When combined with the credit channel—which ensures that the marginal value of money at the lower bound remains finite ($V'(0) < \infty$)—this incentive leads a substantial measure of households to optimally forgo cash holdings and rely on credit. Meanwhile, liquidity-rich households face lower marginal utility, allowing them to maintain large balances to capture deposit returns.

Under these conditions, the presence of banking services leads to a significant reduction in equilibrium money demand—measured by average real balances—and drives the stationary distribution toward a more pronounced right-skewed shape. Figures 6a and 6b summarize these implications by comparing the distribution of real balances (z) in the baseline banking economy versus the self-insurance benchmark under a 3.1% inflation stationary equilibrium, which corresponds to the average U.S. inflation rate over our calibration period (1983-2007).

In the CDF (Figure 6a), the banking-economy curve lies strictly above its no-bank counterpart,

indicating that for any given threshold z , a larger fraction of households holds at most z when banking services are available. Notably, the discrete jump at $z = 0$ captures the positive mass of hand-to-mouth households whose emergence is endogenously driven by credit access. The PDF (Figure 6b) reinforces this message from a density perspective: probability mass increases sharply near zero balances and declines over intermediate ranges, while a thin but long right tail of depositors remains. The availability of banking shifts the overall distribution of liquidity toward the left and generates an HTM mass, a result that is consistent with the empirical characteristics of liquid asset distributions (Figure 1).

Figure 6: Effect of banking on money holding distribution.

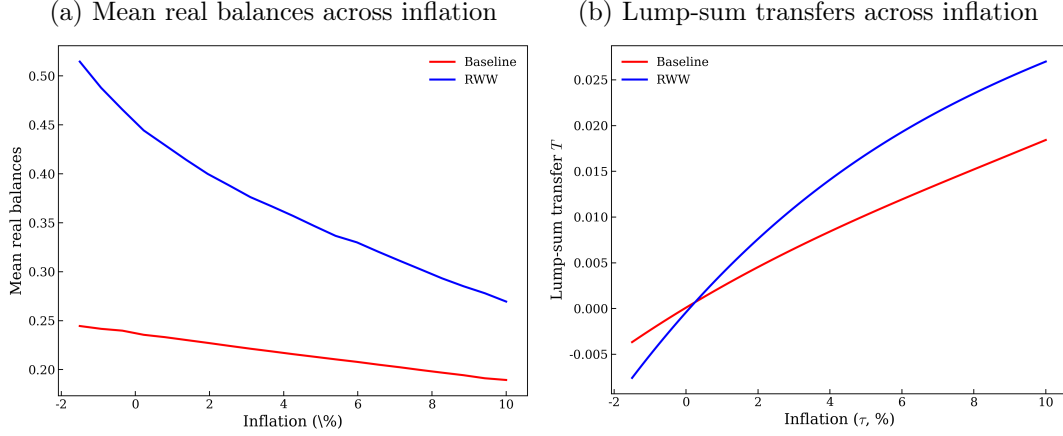


Consequently, the presence of banking leads to a reduction in aggregate money demand, which directly scales down the equilibrium lump-sum transfer. Given that the transfer is proportional to the inflation tax base, a smaller aggregate stock of real balances necessarily implies smaller transfers in absolute value. Figure 7 quantifies this relationship across various inflation rates. At every level of inflation, mean real balances are lower in the banking economy than in the self-insurance benchmark, a result shown directly in Figure 7. The behavior of transfers follows this contraction in the tax base, though the qualitative impact varies depending on whether the inflation rate is positive or negative—reflecting the transition between a lump-sum rebate and a lump-sum tax.

3.2 Mechanisms: Insurance and Distributional Effects

Competitive banking affects allocations through two channels: an insurance effect and a distributional effect. The insurance effect captures how banking alters households' choices conditional on initial real balances z : credit allows liquidity-poor households to smooth DM consumption by borrowing when a trading opportunity arrives, while deposit services raise the return on carrying liquidity forward and thereby affect saving and CM consumption. The distributional effect captures how these state-contingent incentives reshape the stationary distribution of real balances—

Figure 7: Effect of banking on real balances and lump-sum transfers.



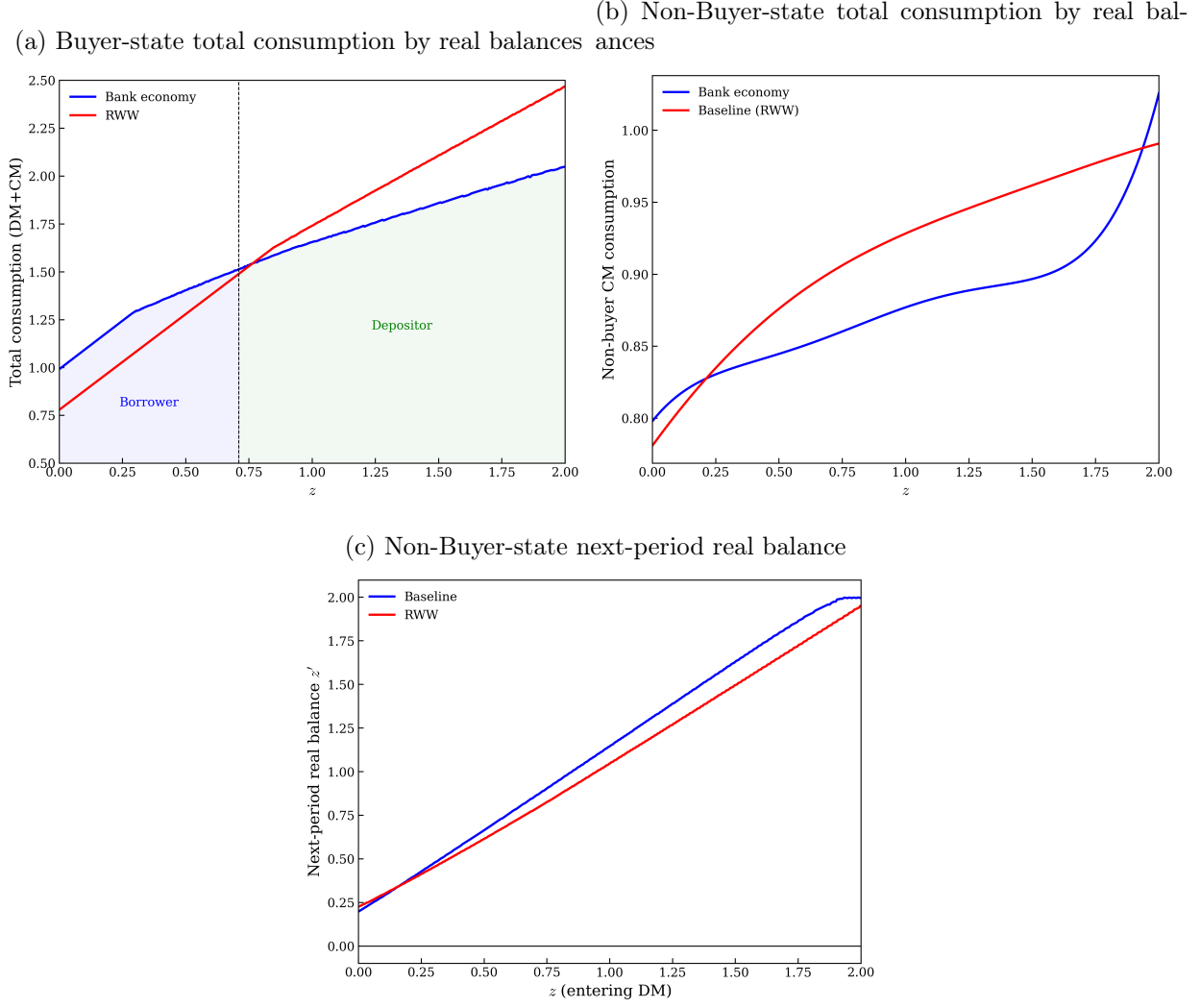
--changing the prevalence of low- versus high-liquidity states---and thus reweighting which parts of the consumption and utility schedules receive the most mass in aggregates and welfare. We now describe these channels in turn, starting with the insurance effect and then turning to the distributional effect.

Insurance effect. In the DM-trade state (probability α), the insurance effect is highly concentrated at the lower tail of the distribution, where banking lifts consumption significantly more for liquidity-poor households than for those with higher balances. For households at the lower end of the liquidity distribution, the no-bank benchmark presents a binding liquidity constraint, necessitating self-insurance through costly cash holdings; competitive banking relaxes this constraint by providing credit, which directly increases DM consumption. Conversely, for liquidity-abundant households, the marginal insurance benefit of credit is minimal as they are already unconstrained in the benchmark economy. For these households, banking primarily influences the intertemporal margin: the incentive to earn deposit returns can shift resources from current consumption to future saving (z'), while the reduction in equilibrium lump-sum transfers—stemming from lower aggregate money demand—further dampens their total consumption.

Figure 8a captures this asymmetry by plotting total buyer-state consumption as a function of z . The vertical cutoff separates the borrower region from the depositor region. In the borrower region, the banking schedule lies markedly above the no-bank benchmark, underscoring the substantial welfare gain from credit access. In the depositor region, the consumption gap narrows and can even turn negative, reflecting the dominant influence of the saving response and reduced transfer resources.

For depositors (non-buyers), banking can reduce current CM consumption through higher saving and a smaller transfer. In this case, the key margin is intertemporal: how much of current resources households consume in the CM versus how much they carry into the next period as z' . A higher deposit return raises the payoff from carrying liquidity forward, improving insurance in a forward-looking sense, but it need not raise CM consumption pointwise in z . In particular, for high- z households, the saving response can dominate: they optimally shift resources into next-period balances and consume less today, and a smaller transfer further compresses contemporaneous CM resources.

Figure 8: Insurance effect: consumption



Figures 8b and 8c make this mechanism transparent. Sub-figure 8b plots non-buyer CM consumption against z . To avoid confusion, note that the CM policy functions are naturally written in terms of the net position $d - l$, whereas the mechanism figures below index outcomes by the initial DM real balances z . In the non-trade state, $l = 0$, so $d - l = z$ and using z as the horizontal axis is without loss of generality. At low z , banking can raise CM consumption because deposit services relax scarcity: earning interest on limited balances meaningfully improves effective resources. At high z , however, CM consumption under banking can lie below the no-bank schedule even though deposit interest is present. Sub-figure 8c explains why: the bank economy features a higher saving choice z' in the non-trade state, especially at higher z . Since CM consumption is what remains after choosing next-period balances, a larger z' mechanically lowers current CM consumption. In addition, the smaller equilibrium rebate implied by lower money demand can further compress CM resources at high z , reinforcing the same pattern.

Taken together, the insurance effect implies that welfare gains from banking can be dominated by borrower-side improvements, while depositor-side gains can be limited in a heterogeneous-agent

equilibrium. In the trade state, the main gain comes from credit relaxing DM constraints at low z . In the non-trade state, deposit services primarily operate by changing intertemporal choices: households may respond by saving more (carrying higher z') rather than by consuming more today, and the smaller equilibrium rebate associated with compressed money demand can further dampen high- z consumption.

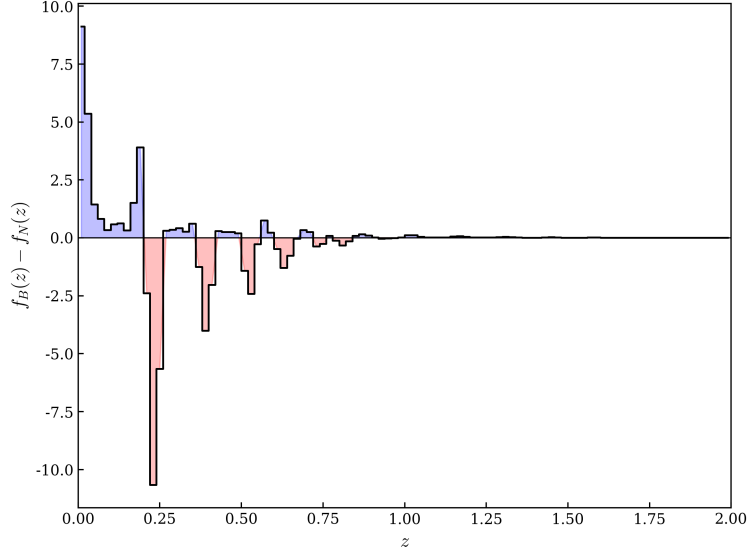
Distributional effect. On the other hand, in a heterogeneous-agent environment, while the insurance-driven increase in consumption for a given level of real balances (z) is important, it is equally critical to examine the distributional effect, which characterizes how banking reshapes the stationary distribution of liquidity across the population.

The distributional effect captures how banking changes not only decision rules at a given z , but also the stationary distribution of z itself through credit and deposit incentives. This channel matters because aggregate consumption and welfare are averages of z -dependent schedules taken under the endogenous stationary distribution. As a result, pointwise insurance gains do not mechanically translate into higher aggregates: even if banking raises consumption at each z over some range, it can lower aggregate consumption if it shifts enough probability mass toward low balances where consumption levels are intrinsically lower. In our model, credit access reduces the need for self-insurance and creates a pile-up near zero balances, while deposit services reshape saving incentives across the distribution; together these forces reallocate stationary mass toward the lower tail. This reweighting changes which part of the consumption and utility schedules matters quantitatively---potentially amplifying borrower-side gains where banking relaxes tight constraints, but also reducing the weight on high-balance regions where consumption is high and where banking may mainly induce more saving or smaller CM resources.

Figure 9 visualizes the distributional effect by plotting the density difference $f_B(z) - f_N(z)$, i.e., the banking-economy PDF minus the no-bank PDF at each level of real balances z . This object has a direct interpretation: where $f_B(z) - f_N(z) > 0$, banking makes that balance level more prevalent in the stationary cross-section (more probability mass under banking than under self-insurance); where $f_B(z) - f_N(z) < 0$, banking makes that region less prevalent. The figure shows a positive difference near low z , consistent with an endogenous hand-to-mouth mass and a thicker lower tail under banking, and a negative difference over intermediate and higher ranges, indicating that banking shifts mass away from those balance levels. This reweighting is quantitatively important because aggregates and welfare are averages of z -dependent consumption and utility schedules: changing where the stationary mass lies can tilt aggregate outcomes toward the regions that receive more weight under banking, even when the insurance schedules themselves cross across z .

Summary. This logic clarifies why our heterogeneous-agent results can differ from BCW: in our model, welfare gains can be driven primarily by borrowers through credit, while depositors can see smaller gains or even welfare losses, and these asymmetric effects are further shaped by distributional reweighting. In BCW's representative-agent environment, the distributional channel is absent--there is no cross-sectional reweighting across liquidity states--so deposit services summarize the insurance benefit at the aggregate level mainly by raising the return to carrying liquidity. In our heterogeneous-agent economy, by contrast, the insurance effect is inherently balance-dependent: credit delivers large welfare gains at the lower tail by relaxing DM constraints, whereas for high- z

Figure 9: Distributional effect



households deposit services mainly operate through stronger saving incentives (and potentially smaller equilibrium CM resources), which can limit or reverse the pointwise welfare gain. The distributional effect then magnifies the borrower-driven nature of aggregate welfare by shifting stationary mass toward low- z regions where credit insurance is most valuable and away from high- z regions where net gains are weakest.

3.3 Welfare implications of banking

In this section, we study the welfare implications of banking, focusing on the trade-off between the two channels highlighted in Section 3.2: an insurance effect---how banking changes allocations at a given liquidity state---and a distributional effect---how banking changes the stationary prevalence of those liquidity states. Inflation matters because it reshapes both channels and, crucially, it changes which households drive the aggregate welfare gain.

Welfare is evaluated at z , the real balances a household carries into the DM at the beginning of the period. In economy $j \in \{B, N\}$, define the within-period expected utility induced by the optimal policies as

$$u_j(z) \equiv \alpha \left[v(y_j(z)) + u(c_j(z)) - \bar{h} \right] + (1 - \alpha) \left[u(\tilde{c}_j(z)) - \bar{h} \right]. \quad (3.10)$$

While CM policy functions are naturally written as functions of the net bank position $x \equiv d - l$ (see Section 2), in this section we index allocations by the initial DM real balance z using the equilibrium mapping from Section 2. Aggregate welfare is the stationary expectation of $u_j(z)$,

$$W_j = \int u_j(z) dF_j(z). \quad (3.11)$$

The welfare gain from banking is

$$\Delta W \equiv W_B - W_N. \quad (3.12)$$

To separate the welfare gain from banking into an insurance effect and a distributional effect, we use the following reference-distribution decomposition. For notational convenience, define $\mathcal{W}(u, F) \equiv \int u(z) dF(z)$, so that $W_j = \mathcal{W}(u_j, F_j)$. Anchoring the decomposition at the banking-economy distribution F_B , we write

$$\Delta W = \left[\mathcal{W}(u_B, F_B) - \mathcal{W}(u_N, F_B) \right] + \left[\mathcal{W}(u_N, F_B) - \mathcal{W}(u_N, F_N) \right], \quad (3.13)$$

and refer to the first bracket as the insurance effect (policy at fixed F_B) and the second bracket as the distributional effect. The term $\mathcal{W}(u_N, F_B)$ is a counterfactual evaluation of the no-bank utility schedule under the bank-economy stationary distribution. It is introduced solely for decomposition: it does not correspond to an equilibrium of the no-bank economy under F_B .

The insurance effect isolates how access to credit and deposits changes within-period utility pointwise in z holding fixed the cross-sectional distribution of liquidity, while the distributional effect captures how banking shifts the stationary distribution of liquidity and thereby reweights which parts of the no-bank utility schedule matter for aggregate welfare.

After presenting the baseline decomposition anchored at the banking-economy distribution F_B , it is useful to note that an alternative symmetric decomposition can be constructed by anchoring at the no-bank distribution F_N . In particular, we can equivalently write

$$\Delta W = [W(u_B, F_N) - W(u_N, F_N)] + [W(u_B, F_B) - W(u_N, F_B)] \quad (3.14)$$

In this formulation, the first term measures the insurance effect evaluated at the no-bank distribution, while the second term captures the distributional reweighting evaluated under the banking utility schedule. Quantitatively, our results are robust to this alternative anchoring: the insurance component remains the dominant positive force, while the distributional component remains negative but shrinks in magnitude as inflation rises. We therefore focus on the F_B -anchored decomposition for expositional clarity.

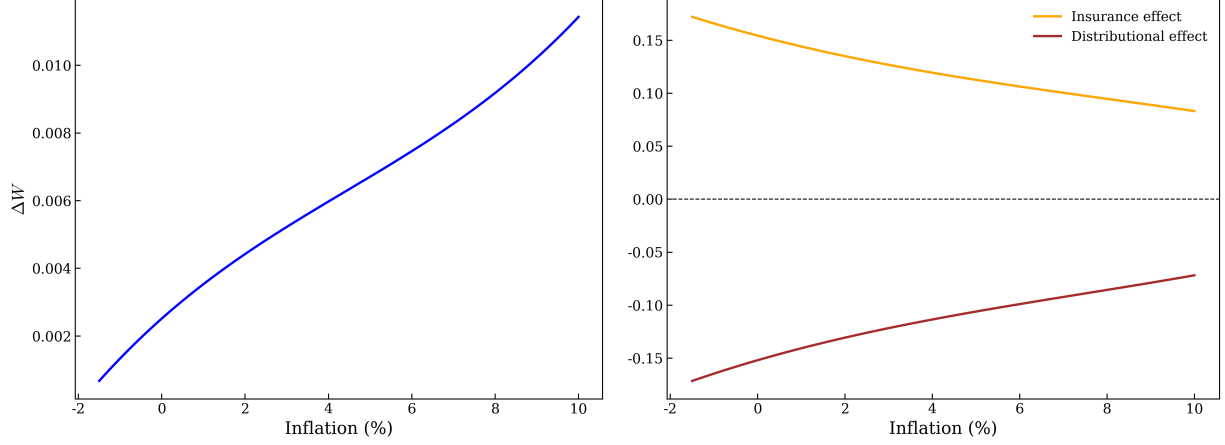
Our quantitative results indicate that banking remains essential even in the heterogeneous-agent economy. Panel (10a) plots the welfare gain from banking across inflation rates ranging from near the Friedman rule to 10 percent. The welfare gain increases with inflation over this range.

Decomposing the welfare gain reveals a clear pattern: the insurance effect is the main driver of welfare gains from banking, but it declines with inflation, whereas the distributional effect remains negative but becomes less adverse as inflation rises. Figure 10b summarizes this decomposition. The figure shows that the insurance effect stays positive but decreases with inflation, while the distributional effect stays negative but increases toward zero as inflation rises.

The insurance effect is largest at low inflation because borrowing costs are low, so households make more intensive use of credit and the associated consumption-smoothing gain is strong. As inflation rises, the insurance effect declines because higher interest rates raise borrowing costs and

Figure 10: Aggregate welfare effects of banking: Insurance and Distributional Effects

(a) Aggregate welfare gain from banking across inflation (b) Decomposition of the welfare gain into Insurance and Distributional Effects



weaken the incremental increase in consumption financed by credit. In addition, higher deposit returns strengthen saving incentives, which can shift resources toward future balances and reduce contemporaneous consumption, further dampening the measured insurance gain.

The distributional effect is negative, but it increases with inflation (i.e., becomes less adverse). The reason is that higher inflation pushes both economies toward lower money holdings, so the stationary distributions F_B and F_N become more similar as inflation rises. When inflation is low, banking substantially compresses money demand relative to the self-insurance benchmark, generating a sizable reallocation of stationary mass toward the lower tail; because low- z states are associated with lower consumption and higher marginal utility, this reweighting is quantitatively important and shows up as a sizable negative distributional term when evaluated on the no-bank utility schedule. As inflation increases, however, self-insuring households also economize on cash holdings and the no-bank distribution shifts left on its own, leaving less scope for banking to further reshape the cross-sectional distribution. Consequently, the cross-economy distributional gap shrinks and the negative reweighting term mechanically moves toward zero.

To sharpen the mechanism behind these aggregate patterns, we further decompose welfare into borrower and depositor components. To compare across economies, we use a common partition of the DM state space based on initial balances z . Let $I(z) \in \{0, 1\}$ indicate whether a household with DM balances z borrows when a DM trading opportunity arrives (and apply the same cutoff rule in both economies so that the partition is comparable). Define borrower-side welfare as the buyer-state contribution integrated over the borrowing region,

$$W_j^{\text{borr}} \equiv \alpha \int I(z) \left[v(y_j(z)) + u(c_j(z)) - \bar{h} \right] dF_j(z). \quad (3.15)$$

Define depositor-side welfare as the complementary contribution (the non-borrowing region in

the buyer state plus all non-trade realizations),

$$W_j^{\text{dep}} \equiv \alpha \int (1 - I(z)) \left[v(y_j(z)) + u(c_j(z)) - \bar{h} \right] dF_j(z) + (1 - \alpha) \int \left[u(\tilde{c}_j(z)) - \bar{h} \right] dF_j(z). \quad (3.16)$$

The corresponding welfare gains are

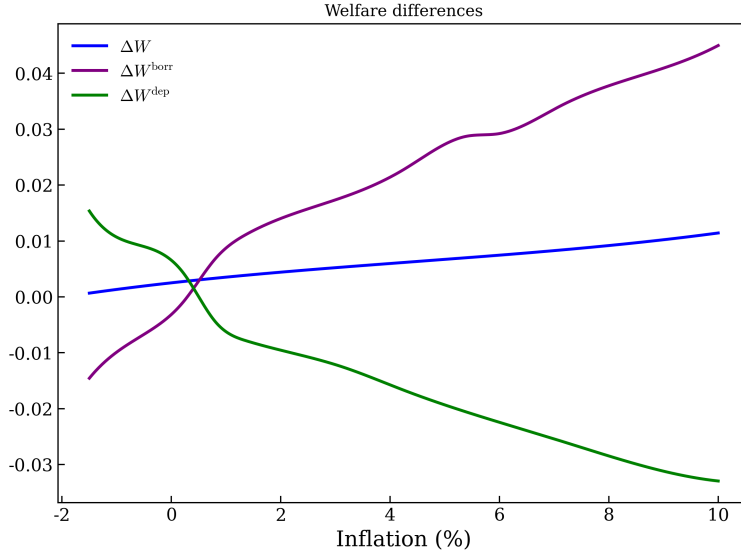
$$\Delta W^{\text{borr}} \equiv W_B^{\text{borr}} - W_N^{\text{borr}}. \quad (3.17)$$

$$\Delta W^{\text{dep}} \equiv W_B^{\text{dep}} - W_N^{\text{dep}}. \quad (3.18)$$

By construction, $\Delta W = \Delta W^{\text{borr}} + \Delta W^{\text{dep}}$.

Using this type decomposition, we next examine how borrower- and depositor-side welfare gains vary with inflation. We begin with borrowers. Figure 11 plots the borrower contribution to the aggregate welfare gain, ΔW^{borr} , across inflation rates. The figure shows a clear monotonic pattern: the borrower contribution is negative at low inflation, but rises sharply with inflation and eventually becomes strongly positive.

Figure 11: Welfare Decomposition: Borrower and Depositor

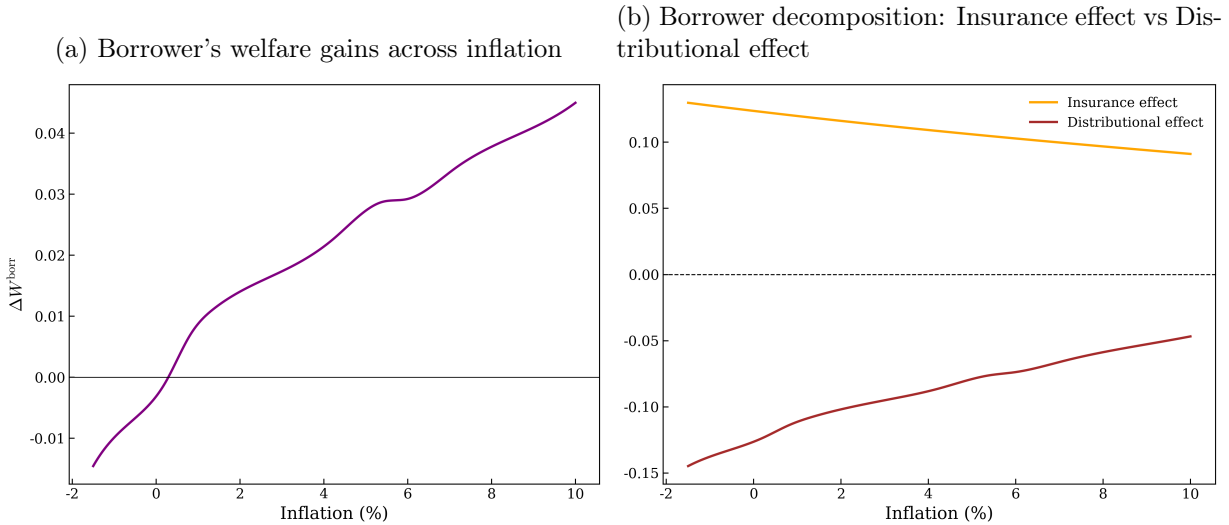


This borrower-driven pattern also highlights an important difference from BCW. In BCW's representative-agent environment, welfare gains from banking operate primarily through deposit services: higher inflation raises the return advantage of deposits relative to money and thereby increases welfare through improved insurance. In our heterogeneous-agent economy, by contrast, the aggregate welfare gain becomes increasingly borrower-driven: as inflation rises, the decisive margin is credit-based consumption smoothing for liquidity-poor households, while depositor-side gains are limited and can turn negative once saving incentives and general-equilibrium resource adjustments are taken into account. To understand why borrowers switch from losses to gains

as inflation rises, we now decompose the borrower contribution into insurance and distributional components.

The borrower contribution rises with inflation because the distributional headwind fades. Figure 11 decomposes ΔW^{borr} into insurance and distributional effects. Two features stand out. First, the borrower insurance component is positive but gently declines with inflation, consistent with DM trade being compressed and borrowing being used less intensively at a given z as borrowing costs rise. Second, the decisive movement comes from the distributional component: at low inflation, banking induces a sizable reallocation of stationary mass that shows up as a large negative distributional term; as inflation rises, the no-bank distribution shifts toward low balances and becomes closer to the bank distribution, so this distributional penalty collapses. Once this headwind fades, the positive borrower insurance gain dominates, turning the borrower contribution positive and increasing.

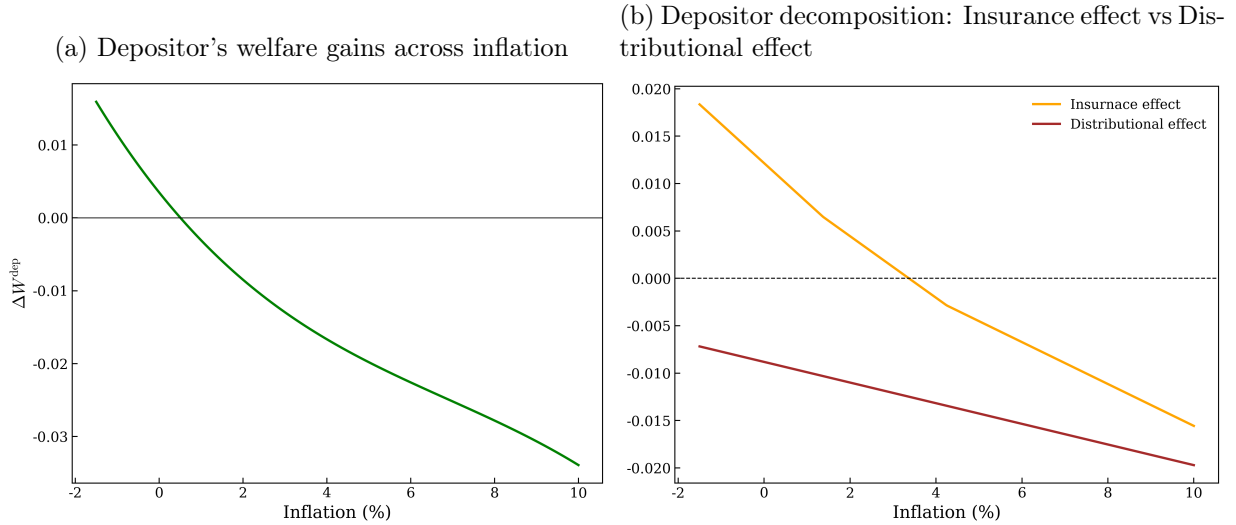
Figure 12: Borrower-side welfare effects and decomposition: Insurance and Distributional Effects



Depositor-side welfare moves in the opposite direction: it is positive at low inflation but declines with inflation and eventually turns negative. Figure 13a documents this monotone decline in ΔW^{dep} . The decomposition in Figure 13b clarifies why. First, the depositor insurance component falls with inflation because deposit services are increasingly absorbed by the saving margin. As the deposit return rises, high-balance households optimally reallocate resources toward next-period liquidity (higher z') rather than toward contemporaneous CM consumption, so within-period utility gains shrink and can turn negative. Second, the depositor distributional component remains negative and becomes more adverse because banking compresses the upper tail of real balances: relative to the no-bank benchmark, the stationary distribution places less weight on high- z states under banking, exactly the region where depositors live and where within-period utility levels are high. This reweighting therefore reduces depositor-side welfare when evaluated in the cross-section, and the effect strengthens with inflation because the saving response that pulls mass away from high balances becomes more pronounced. Changes in the equilibrium lump-sum resource component associated with lower aggregate money demand can reinforce these depositor losses,

but quantitatively it is secondary to the saving-driven decline in within-period utility and the unfavorable reweighting of high- z states.

Figure 13: Depositor-side welfare effects and decomposition: Insurance and Distributional Effects



Summary. Banking remains welfare-relevant in the heterogeneous-agent economy, and the aggregate welfare gain increases with inflation over the range from near the Friedman rule to 10 percent. Decomposing the welfare gain shows that the insurance effect is the primary source of gains, but it declines with inflation as higher borrowing costs compress credit use and stronger deposit returns are increasingly absorbed by saving rather than contemporaneous consumption. The distributional effect is negative---banking shifts the stationary distribution of real balances toward lower liquidity states---but it becomes less adverse as inflation rises because both economies converge to low money holdings and the cross-economy distributional gap shrinks.

A borrower--depositor decomposition reveals that these aggregate patterns mask sharply different forces across the distribution. Borrowers contribute negatively at low inflation but increasingly positively at higher inflation, because the negative distributional headwind that dominates at low inflation collapses rapidly with inflation, leaving the (still positive) borrower insurance gain to dominate. Depositors move in the opposite direction: their contribution is positive at low inflation but declines and turns negative as inflation rises. This occurs because higher deposit returns primarily raise saving rather than within-period utility, and because banking compresses the upper tail of real balances, reducing stationary weight on high- z states where depositor welfare is high. Overall, unlike BCW where welfare gains operate mainly through deposit services in a representative-agent setting, welfare gains in our heterogeneous-agent economy become increasingly borrower-driven as inflation rises.

4 Conclusion

In this paper, we revisit the welfare role of competitive banking in a search-theoretic monetary economy by embedding the BCW banking structure into an empirically relevant heterogeneous-agent environment. Unlike the representative-agent benchmark, our model generates a non-degenerate distribution of liquid assets and an endogenous hand-to-mouth mass: a positive measure of households optimally chooses zero next-period real balances. This endogenous lower-tail behavior is central for understanding how banking reshapes liquidity demand and welfare. In particular, bank credit substitutes for cash self-insurance and compresses equilibrium money demand, even though deposit services raise the return on idle liquidity.

Our quantitative results show that banking remains essential: the welfare gain from introducing competitive banking is positive under our U.S. calibration and increases with inflation over the policy range we consider. However, the source and composition of these gains differ sharply from the representative-agent logic emphasized in BCW. In our heterogeneous economy, banking affects welfare through two distinct channels. The insurance effect captures how borrowing and deposit opportunities change allocations conditional on a given liquidity state, while the distributional effect captures how banking reshapes the stationary distribution of liquidity and thereby reweights which states matter for aggregate welfare. This second channel is absent in representative-agent environments and is quantitatively important in our setting.

A key implication is that the incidence of banking’s welfare benefits is inflation-dependent. At low inflation, welfare gains are relatively more deposit-driven, reflecting improvements in the non-trade state through deposit returns and intertemporal insurance. As inflation rises, the dominant source of welfare gains shifts toward borrowers: higher inflation increases the opportunity cost of carrying cash, making credit a more valuable substitute for self-insurance, while the adverse distributional headwind from banking-induced reweighting becomes less pronounced as the self-insurance economy also economizes on liquidity. Overall, the model delivers an empirically plausible right-skewed liquidity distribution with a hand-to-mouth mass and shows that competitive banking’s welfare relevance cannot be inferred from representative-agent logic alone.

These findings suggest that evaluating banking and monetary policy jointly requires tracking the endogenous distribution of liquidity. In economies where liquidity is unevenly held and hand-to-mouth behavior is an equilibrium outcome, banking can simultaneously reduce money demand, change the inflation-tax base, and alter the cross-sectional incidence of inflation and transfers. More broadly, our results highlight that the welfare consequences of financial intermediation depend not only on pointwise insurance gains, but also on how intermediation reshapes the distribution of liquid assets—and that this interaction can fundamentally change which households drive the welfare gains from banking across inflation regimes.

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A Model Appendix

The appendix is organized as follows: Section A.1 establishes the technical properties of the model, including interior borrowing and the finiteness of $V'(0)$. Section A.2 explains the numerical algorithm used to solve the model. Section A.3 describes the calibration.

A.1 Technical Properties of the Model: Interior Borrowing and the Finiteness of $V'(0)$

This section formalizes the conditions under which borrowing is interior at the lower bound of real balances and the marginal value of money at zero is finite. These properties are central for the existence of an endogenous hand-to-mouth mass in the banking economy.

Lemma 3. *Suppose preferences satisfy $v' > 0$, $v'' < 0$, $u' > 0$, and $u'' < 0$, with $v'(0) = +\infty$ and $u'(c) < \infty$ for all $c > 0$. Let $\bar{l}$ denote the borrowing limit. If there exists $l^* > 0$ satisfying*

$$v'(l^*) = (1+i)u'(\bar{h} + T - (1+i)l^*)$$

and $l^ < \bar{l}$, then borrowing is interior at $z = 0$.*

Proof. Consider a household entering the DM with $z = 0$. Upon receiving a consumption opportunity, the household solves

$$\max_{0 \leq l \leq \bar{l}} v(l) + W(-l, l, 0).$$

Using the envelope condition $W_l = -(1+i)u'(c)$ from the CM problem, the first-order condition for an interior solution is

$$v'(l) - (1+i)u'(c) = 0,$$

where $c = \bar{h} + T - (1+i)l$ denotes CM consumption. Because $v'(0) = +\infty$ and $u'(\cdot)$ is finite for strictly positive consumption, the objective is strictly increasing at $l = 0$. Hence, $l = 0$ cannot be optimal. Strict concavity of the objective implies a unique interior solution $l^* > 0$ satisfying

$$v'(l^*) = (1+i)u'(\bar{h} + T - (1+i)l^*).$$

If $l^* < \bar{l}$, the borrowing constraint does not bind, and borrowing is interior at $z = 0$. □

Lemma 4. *Under the conditions of Lemma 3 and assuming interior borrowing at $z = 0$, the marginal value of money at the lower bound is finite.*

Proof. The marginal value of money at the beginning of the DM is

$$V'(z) = \alpha \cdot v'(y(z)) + (1 - \alpha)(1 + i)u'(\tilde{c}(z))$$

Evaluating at $z = 0$ and using interior borrowing $l(0) = l^* > 0$ yields

$$V'(0) = \alpha v'(l^*) + (1 - \alpha)(1 + i)u'(\bar{h} + T - R^{-1}z').$$

Since $l^* > 0$ and preferences are strictly concave, both $v'(l^*)$ and $u'(\cdot)$ are finite. Therefore

$$V'(0) < \infty.$$

By contrast, in the self-insurance economy without credit, households must finance DM consumption entirely with balances. Because $v'(0) = +\infty$, the marginal value satisfies

$$V'(0) = +\infty,$$

which precludes hand-to-mouth behavior in stationary equilibrium. $\square$

Corollary 1. *Suppose the conditions of Lemma 3 (interior borrowing at $z = 0$) and Lemma 4 (finiteness of $V'(0)$) hold. Consider the CM problem with state $x \equiv d - l$ (net deposit) and choice $z' \geq 0$. Assume u is strictly concave and satisfies $u'(0) = +\infty$ and $u'(\infty) = 0$, and V is concave with $V'(0) < \infty$. Then there exists a cutoff $\tilde{l} \in \mathbb{R}$ such that the optimal CM saving policy satisfies*

$$z'(x) = 0 \quad \text{for all } x \leq \tilde{l}, \quad z'(x) > 0 \quad \text{for all } x > \tilde{l}.$$

Equivalently, there exists a non-empty set of states for which households optimally choose to carry zero next-period real balances and hence behave hand-to-mouth.

Proof. Let x denote CM net resources after DM and bank settlement (net deposit position). The CM problem can be written as

$$W(x) = \max_{c \geq 0, z' \geq 0} \{u(c) - \bar{h} + \beta V(z')\} \quad \text{s.t.} \quad c + R^{-1}z' \leq (1 + i)x + \bar{h} + T.$$

Substituting the constraint yields the equivalent problem

$$\max_{z' \geq 0} u(x + \bar{h} + T - R^{-1}z') + \beta V(z').$$

Define the objective $\Phi(z'; x)$. By strict concavity of u and concavity of V , $\Phi(\cdot; x)$ is strictly concave

in z' , implying a unique optimizer. The derivative at the boundary $z' = 0$ is

$$\partial_{z'}\Phi(0; x) = -R^{-1}u'((1+i)x + \bar{h} + T) + \beta V'(0).$$

Since $V'(0) < \infty$, there exists a unique cutoff $\tilde{l}$ solving

$$R^{-1}u'((1+i)\tilde{l} + \bar{h} + T) = \beta V'(0).$$

For $x \leq \tilde{l}$, we have $u'((1+i)x + \bar{h} + T) \geq u'((1+i)\tilde{l} + \bar{h} + T)$, so $\partial_{z'}\Phi(0; x) \leq 0$, and the unique optimum is the corner solution $z'(x) = 0$. For $x > \tilde{l}$, $\partial_{z'}\Phi(0; x) > 0$, and strict concavity implies an interior solution satisfying the Euler condition with $z'(x) > 0$. Hence, the hand-to-mouth region $\{x : z'(x) = 0\}$ is non-empty and characterized by the cutoff $\tilde{l}$. $\square$

Proposition 1. *Maintain the conditions of Corollary ???. Let G denote the stationary cross-sectional distribution of net deposit positions x at the beginning of the CM. Assume G admits a density g that is strictly positive on an interval containing $\tilde{l}$. Then, in the stationary monetary equilibrium,*

$$\Pr(z' = 0) = G(\tilde{l}) > 0,$$

so a positive measure of households is hand-to-mouth.

Proof. By Corollary ??, $z'(x) = 0$ for all $x \leq \tilde{l}$. Therefore

$$\Pr(z' = 0) = \Pr(x \leq \tilde{l}) = G(\tilde{l}).$$

Since $g(x) > 0$ on an interval containing $\tilde{l}$, there exists $\varepsilon > 0$ such that $g(x) > 0$ for all $x \in [\tilde{l} - \varepsilon, \tilde{l}]$. Hence

$$G(\tilde{l}) \geq \int_{\tilde{l}-\varepsilon}^{\tilde{l}} g(x) dx > 0.$$

Thus, a positive measure of households chooses $z' = 0$ in stationary equilibrium. $\square$

A.2 Numerical algorithm for finding a SME

The algorithm to compute the value functions, policy functions and distribution of real balances at a stationary equilibrium is composed of following steps:

1. Given the money growth rate τ , choose the initial nominal interest rate, i , and monetary transfer, T . For example, $i = \tau$ and $T = 0$.

2. Guess the initial value of $V_0(z)$ for the domain $z \in [0, \bar{z}]$ where z is the amount of real balance at the start of the DM and $\bar{z}$ is a sufficiently large number.
3. Guess the initial value of $W_0(A)$ for the domain $A \in [-\bar{l}, \bar{z}]$ where A is net deposit, $d - l$, at the start of the CM.
4. Iterate the following k-th value functions

$$\begin{aligned} V_k(z) &= \alpha \max_{y,A} \{v(y) + W_k(A)\} + (1 - \alpha)W_k(z) \\ W_k(A) &= \max_{c,z'} \{u(c) - \bar{h} + \beta V_{k+1}(z')\} \end{aligned}$$

until both value functions converge, for example, $\max(\|V_{k+1} - V_k\|, \|W_{k+1} - W_k\|) \leq 10^{-5}$. Obtain the stationary value functions, $V(z)$ and $W(A)$.

5. Obtain the CM policy functions, $c(A)$ and $z'(A)$, and DM policy functions, $l(z)$, $d(z)$ and $y(z)$.
6. Initialize a distribution over DM real balances on the grid $\{z_m\}_{m=1}^{N_z}$. Let $F^{(0)} = (F_1^{(0)}, \dots, F_{N_z}^{(0)})$ denote the initial probability mass function (PMF) on this grid, with $F_m^{(0)} \geq 0$ and $\sum_{m=1}^{N_z} F_m^{(0)} = 1$ (e.g., uniform mass on $[0, \bar{z}]$ or a point mass).
7. Given DM policies $(l(\cdot), d(\cdot))$ and the CM savings policy $z'(\cdot)$ defined on the net-position grid $x \equiv d - l \in [-\bar{l}, \bar{z}]$, construct the one-step transition implied by the idiosyncratic DM shock. For each DM grid point z_m , define the CM net position under the α -shock (i.e., the buyer shock) as

$$x^\alpha(z_m) \equiv d(z_m) - l(z_m),$$

and under the $(1 - \alpha)$ -shock (i.e., the non-buyer shock) as

$$x^{1-\alpha}(z_m) \equiv z_m.$$

Let $z'(x)$ denote the CM savings policy mapping the CM net position x into next-period DM real balances. Then the implied next-period DM balances are

$$\hat{z}_m^\alpha \equiv z'(x^\alpha(z_m)), \quad \hat{z}_m^{1-\alpha} \equiv z'(x^{1-\alpha}(z_m)) = z'(z_m).$$

Given the current PMF $F^{(k)} = (F_1^{(k)}, \dots, F_{N_z}^{(k)})$ on the grid $\{z_m\}_{m=1}^{N_z}$, update the distribution by pushing forward mass through these two mappings using linear interpolation on the z -grid:

$$F^{(k+1)} = \alpha \mathcal{T}(\{\hat{z}_m^\alpha\}_{m=1}^{N_z}, F^{(k)}) + (1 - \alpha) \mathcal{T}(\{\hat{z}_m^{1-\alpha}\}_{m=1}^{N_z}, F^{(k)}),$$

where $\mathcal{T}(\cdot)$ assigns each mass $F_m^{(k)}$ to the two adjacent grid points bracketing $\hat{z}_m$ (linear interpolation), ensuring $\sum_{m=1}^{N_z} F_m^{(k+1)} = 1$.

8. Iterate Step 7 until convergence, e.g.,

$$\|F^{(k+1)} - F^{(k)}\|_\infty \leq \varepsilon_F,$$

and denote the limiting distribution by F . The associated stationary density on the grid is then given by $f(z_m) \approx F_m/\Delta z$ (or simply use the PMF $\{F_m\}$ for all aggregation and market-clearing integrals).

9. Check if the loan market clears, for example,

$$|\alpha \int d(z) - l(z) dF(z) + (1 - \alpha) \int \tilde{d}(z) dF(z)| \leq 0.01 \cdot \int z dF(z)$$

which means the excess loan (or deposit) is smaller than 1% of the total value of money stock in the economy. If there is excess loan (deposit), then slightly increase (decrease) the nominal interest rate and start again from step 4.

10. Check if the lump-sum monetary transfer satisfies the monetary policy rule. For example,

$$|T - \tau \int z dF(z)| \leq 10^{-5}$$

If it is not satisfied, then set $T = \tau \int z dF(z)$ and start again from step 4.

11. Check if all the markets are cleared.

A.3 Calibration

Functional form and data. We calibrate our model with banking to the U.S. economy. As outlined by [Lagos and Wright \(2005\)](#), the preferences for DM and CM in the model take the form of:

$$v_{DM}(y) = \frac{(y + b)^{1-\sigma} - b^{1-\sigma}}{1 - \sigma} \tag{A.1}$$

$$u_{CM}(c) = B \cdot \log(c) \tag{A.2}$$

with b relatively close to 0. The money demand of the model can be written as

$$\frac{M}{PY} = \frac{\int z dF(z)}{\alpha \int y(z) dF(z) + \bar{h}}. \tag{A.3}$$

The baseline model consists of five parameters: α, β, B, σ and $\bar{h}$. Table 1 presents the parameter values. The model calibration utilizes long-term data covering 1983-2007 to avoid the Great Recession period when interest rates were constrained by the zero lower bound. We use New M1-to-GDP ratio measured by [Lucas and Nicolini \(2015\)](#) for U.S. money demand (M/PY).

External calibration. One period is defined to be annual and β is set to be 0.96, implying 4% annual real interest rate as in the standard literature. The labor supply, $\bar{h}$, is normalized to 1. The average money growth rate is 3.1 % given the sample data.

Table 1: Calibration targets and results

Notation	Parameter	Value	Empirical Targets
External Calibration			
β	Discount factor	0.96	Normalized
$1 + \tau$	Gross money growth rate	1.031	1983-2007 avg. annual inflation, 3.1%
$\bar{h}$	Labor supply	1	Normalized
B	Coeff. on CM consumption	1	Normalized
Internal Calibration			
α	Consumption shock	0.5	Mean of $M/PY = 0.22$
σ	Curv. of DM consumption	0.2	Elasticity of $M/PY = -0.12$

^a The point elasticity refers to the elasticity of M/PY with respect to the nominal interest rate i , evaluated at the data mean of i .

Internal calibration. Then $\Omega = (\sigma, \alpha)$ was chosen such that the model's money demand matches the historical money demand of the U.S from 1983 to 2007. The risk aversion parameter σ influences the elasticity of money demand in our model. As σ increases, households' consumption smoothing motives become stronger, making money demand less elastic with respect to inflation and resulting in a flatter money demand curve. The consumption shock parameter α governs the probability of receiving a consumption opportunity in the DM, which influences the money demand. The result of our internal calibration is presented in Table 2. As shown in Table 2, the model closely matches empirical targets for the elasticity of money demand, money demand ratio (M/PY).

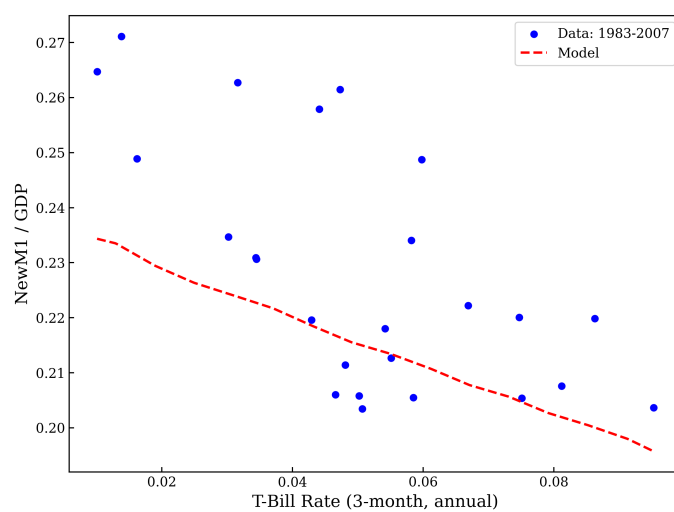
Table 2: Calibration results

Notation	Parameter	Value	Empirical Targets	Model
σ	Curv. of DM consumption	0.2	Elasticity of $M/PY = -0.12$	-0.14
α	Consumption shock	0.5	Mean of $M/PY = 0.22$	0.18

^a The point elasticity refers to the elasticity of M/PY with respect to the nominal interest rate i , evaluated at the data mean of i .

Figure 14 compares the money demand curves from our model and the data. The black line represents the model-generated money demand curve, while the red scatter dots show the empirical observations. As illustrated in Figure 14, the model's predicted money demand appears to fit the data well across the observed range. This close alignment between the model-generated curve and empirical observations provides further validation of our model to capture the key features of money demand behavior in the data.

Figure 14: Money demand: Data and Model



Note: The red dots represent data from 1983 to 2007. The black solid line denotes the money demand curve generated by the model with banking.